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A survey of deep learning-based action quality assessment: methods and applications

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Abstract
Action Quality Assessment (AQA) has recently gained prominence as a crucial area of research, driven by advancements in deep learning and increasing demands for automated, objective evaluations of human action execution. Unlike traditional action recognition tasks, AQA involves a nuanced analysis aimed at quantifying performance quality, thereby enabling practical applications across diverse domains such as sports evaluation, rehabilitation monitoring, and professional skill assessment. This survey systematically reviews deep learning-based methodologies developed for video-based AQA, categorizing existing approaches according to data modalities, learning paradigms, and evaluation granularity. Specifically, the review covers single-modality and emerging multi-modal approaches integrating visual, auditory, and textual information, as well as supervised, self-supervised, and contrastive learning frameworks. Key benchmark datasets utilized in AQA research are comprehensively analyzed with emphasis on their scope, representativeness, and annotation characteristics. Furthermore, critical challenges confronting the field-including limited model generalization, ambiguity in scoring annotations, and interpretability concerns-are identified and discussed. Potential future research directions that could address these limitations and further advance practical AQA deployment are also proposed.

Keywords Action quality assessment, Deep learning, Multi-modal approaches, Action evaluation dataset

Introduction

Action Quality Assessment (AQA) aims to objectively evaluate the quality of human action execution, thus enabling advanced applications in diverse fields, including sports analysis, healthcare, and skill evaluation. Compared with Human Action Recognition (HAR), which primarily concentrates on identifying and classifying different types of actions, AQA involves a more intricate and fine-grained analysis. Specifically, AQA requires systematically examining the entire action sequence to discern subtle variances in execution quality, rather than merely recognizing or categorizing the performed action. Consequently, this inherent complexity renders AQA a considerably more challenging task than traditional HAR.

Over the past decade, Action Quality Assessment (AQA) has attracted increasing attention, leading to multiple surveys that aim to organize the rapidly growing literature. Early reviews, such as Lei et al. [1] and Wang et al. [2], provided broad overviews of AQA methods and benchmark datasets across representative application domains. More recent comprehensive surveys published in 2024-2025 further expanded the coverage and offered systematic summaries of AQA research, including Zhou et al. [3], Liu et al. [4], and Yin et al. [5]. In particular, Yin et al. [5] and Zhou et al. [3] present broader retrospectives over an extended time horizon; however, their coverage predominantly centers on studies published before 2025 and therefore may not fully reflect the most recent methodological advances. Overall, existing reviews have been highly valuable for consolidating the field, yet several gaps remain. First, many surveys place greater emphasis on literature-level aggregation and benchmark-oriented comparisons, while providing comparatively limited synthesis of how deep learning methods evolve along shared modeling threads and how they address recurring challenges (e.g., temporal structure understanding, fine-grained motion reasoning, and uncertainty in subjective scoring). Second, the coverage is often largely vision-centric, whereas recent progress increasingly points to the importance of leveraging complementary cues beyond RGB. Third, systematic discussion from an application-oriented perspective-namely, how assessment requirements, constraints, and desired output forms differ across scenarios and consequently shape modeling choices-remains insufficiently explored.

Recent advancements in multi-modal information fusion have motivated researchers to integrate visual, auditory, and textual modalities, leading to significant improvements in the accuracy and robustness of action quality evaluation, particularly in complex real-world scenarios. Compared to traditional vision-based approaches, multi-modal methods offer richer contextual information and demonstrate greater capability in capturing the intricacies of human action execution. Despite these advancements, current review literature has largely neglected this emerging perspective. Moreover, most existing surveys primarily concentrate on summarizing the performance of AQA methods on specific benchmark datasets-an approach also adopted in the present review. While these methods have achieved notable results on controlled datasets, critical challenges remain in real-world applications. In particular, systematic analysis from an application-oriented perspective has been insufficiently addressed in existing reviews.

To address these gaps, this survey provides a comprehensive and up-to-date review of AQA research from both methodological and application-oriented perspectives. In addition to vision-based pipelines, it explicitly highlights the growing body of work on multi-modal AQA that leverages visual, auditory, and textual cues. The surveyed methods are organized along complementary dimensions, including learning paradigms (e.g., supervised, self-supervised, and contrastive learning), analysis granularity (e.g., holistic scoring versus fine-grained parsing), and the modalities employed. The survey further synthesizes recent progress in multi-modal fusion and discusses how these designs can improve robustness and generalization. Beyond methodological taxonomy, it summarizes practical issues that hinder real-world adoption, such as generalization to unseen subjects and scenarios, label ambiguity and subjectivity, annotation efficiency, and the interpretability of scoring outcomes. Finally, widely used benchmark datasets are revisited with emphasis on their representativeness, diversity, and suitability for different application domains and evaluation settings.

This survey is based on a comprehensive literature collection conducted across major academic databases, including Google Scholar, IEEE Xplore, and Web of Science, which together provide extensive coverage of leading research in computer science, artificial intelligence, and computer vision. To ensure both retrieval accuracy and coverage, targeted keyword combinations such as "Action Quality Assessment," "Deep Learning," and "Computer Vision" were employed using Boolean operators, for example "Action Quality Assessment AND Deep Learning." In addition, synonymous expressions such as "Human Action Evaluation" were incorporated to capture relevant studies adopting alternative terminologies. The search was restricted to publications from 2014 to 2025 in order to reflect recent methodological developments and emerging research trends in the AQA field. The overall literature retrieval and screening workflow is illustrated in Fig. 1.

Following the initial retrieval, a multi-stage screening procedure was applied to identify studies directly relevant to AQA. First, titles, abstracts, and keywords were examined to remove papers that focused exclusively on action recognition, gesture classification, or tasks unrelated to video-based analysis. The remaining studies were then assessed through full-text review, and those that did not treat AQA as a primary research objective or lacked methodological contributions in computer vision, deep learning, or video analysis were excluded. The final set of selected literature was subsequently organized according to two complementary perspectives-application-based classification and modality-based classification-so as to systematically analyze methodological similarities and differences and provide a structured overview of the AQA research landscape.

In summary, with the growing interest and rapid progress in AQA research, there is a clear need for an up-to-date, systematic, and in-depth review that not only captures recent methodological advances-including the emerging trend of multi-modal approaches-but also critically examines their applicability and interpretability in real-world scenarios. By integrating a comprehensive literature screening process with

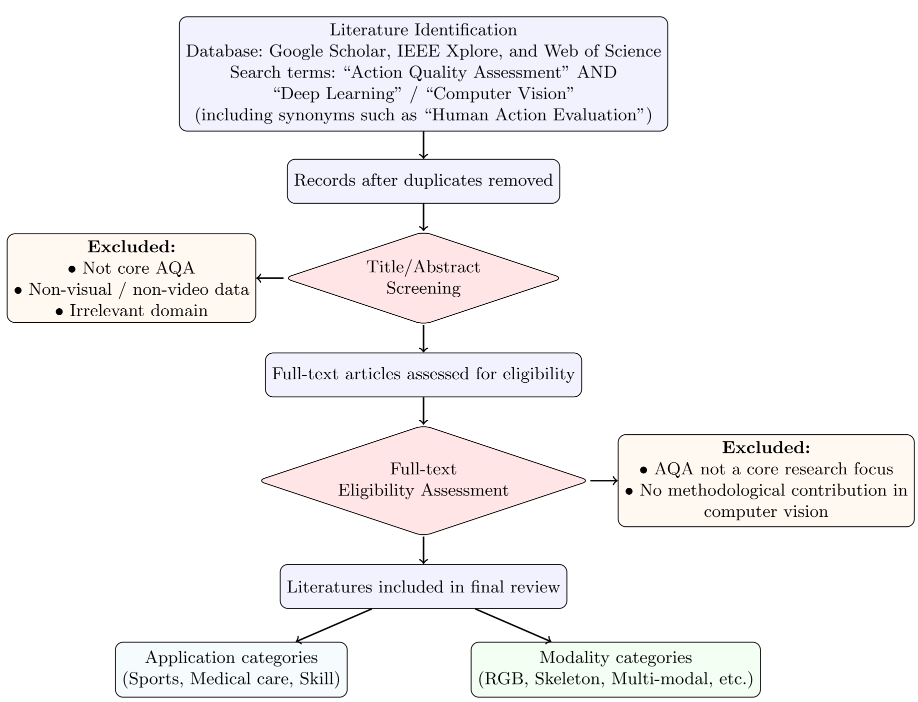

Fig. 1 Literature search, screening, and selection process for AQA studies

structured classification strategies, this review aims to provide a clear overview of the current research landscape, identify existing limitations, and offer valuable insights to guide future investigations and practical implementations in the field of AQA. The main contributions of this paper are summarized as follows:

- A comprehensive review of AQA methods and datasets published in the past 10 years is presented, covering key developments and trends in the field.
- The applications of AQA methods are systematically analyzed across major domains, including sports, medical care, and skill assessment, highlighting representative use cases and practical relevance.
- Current challenges in AQA research are discussed in detail, and potential future directions are proposed to address limitations and promote further advancements, particularly in real-world deployment and multi-modal integration.

The remainder of this paper is structured as follows. Section "Background survey" provides a brief review of the development of AQA methods in recent years and presents a concise introduction to the categorization of the reviewed literature. Section "Datasets and evaluation metrics" presents an overview of AQA datasets from various domains and introduces the commonly used evaluation metrics for AQA tasks. Section "Review of existing AQA methods" offers a detailed introduction to the processes and innovations of various AQA methods. Sections "Application of AQA in sports", "Application of AQA in medical care", and "Application of AQA in skill assessment" sequentially review the applications of AQA in the fields of sports, medical care, and skill assessment. Section "Challenges and future work" discusses the challenges faced by AQA and describes future research directions. Finally, Sect. "Limitation and conclusion" summarizes the content of this review. An overview of the overall structure of this survey is illustrated in Fig. 2.

Background survey

Overview of AQA

Although both AQA and HAR fall under computer vision tasks that involve analyzing human actions in videos, they differ significantly in terms of their objectives, methods, and complexity. HAR consists of two main steps: action detection and classification. First, it detects and segments action sequences in the video, and then classifies them to identify the action type. HAR is relatively simple, as its goal is to distinguish between different actions, where the differences are often more apparent. In contrast, AQA has more refined and complex objectives. Instead of classifying action types, AQA evaluates the quality differences within the same type of action. AQA requires a detailed analysis across various dimensions, such as the fluidity, precision, coordination, and strength control of the action. Figure 3 illustrates the basic architectures commonly used in the AQA and HAR tasks.

With the rapid advancement of deep learning, the potential of AQA is becoming increasingly evident in emerging fields, especially in scenarios where action precision, execution processes, and skill mastery are crucial. AQA technology is primarily focused on three main areas: sports, medical care, and skill assessment. In sports, AQA is widely used to analyze and score athletes' performances [6–9]. It helps coaches and athletes quickly identify areas for improvement and optimize training strategies. In medical care,

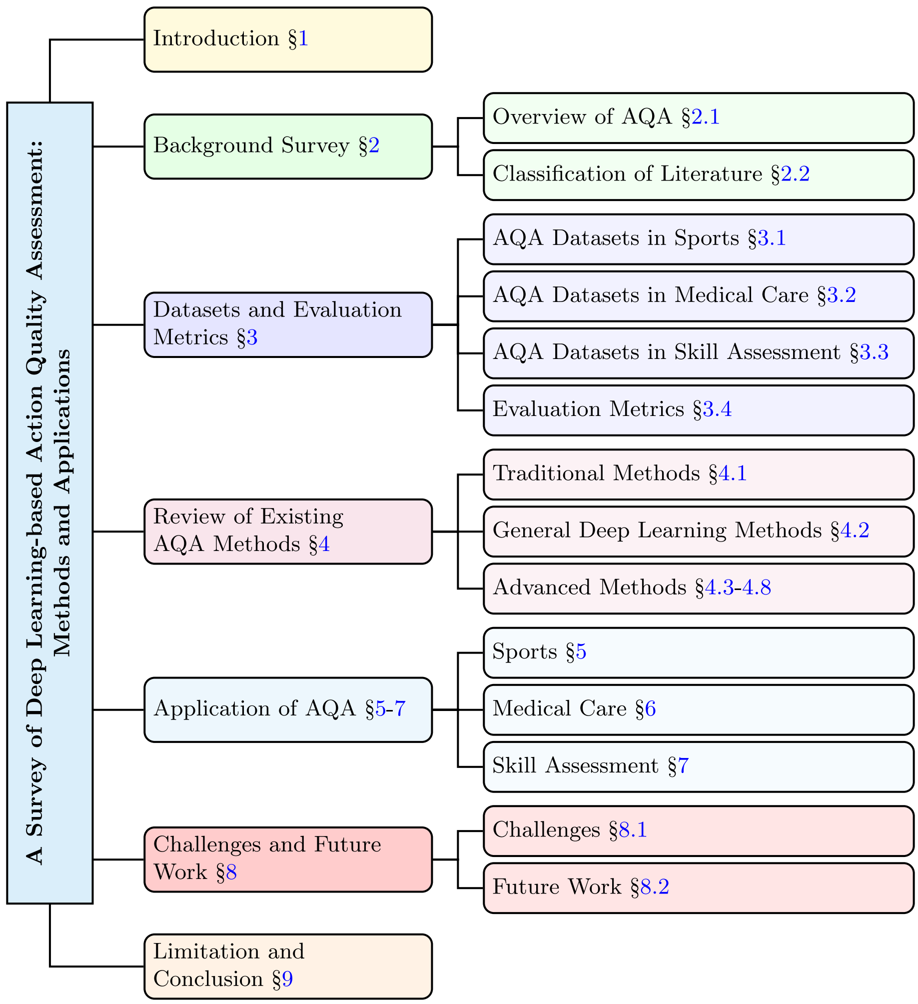

Fig. 2 Overall structure of the survey

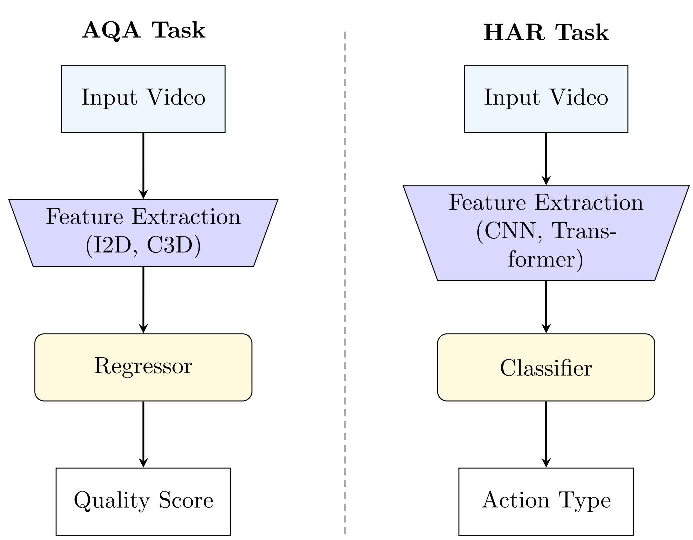

Fig. 3 Comparison between AQA and HAR tasks

AQA can provide accurate assessments of patients' movements to monitor recovery progress and tailor personalized treatment plans [10]. In the area of human skills assessment, AQA is applied to both everyday tasks and professional skills. For example, in daily life, AQA can evaluate the quality of actions in tasks such as braiding hair, tying a tie, and applying eyeliner [11, 12], providing data to improve the efficiency and quality of tasks. AQA can also be applied to skills that require high levels of coordination and precision, such as playing the piano [13], offering automated analysis that provides objective feedback to learners and trainers, thus improving the efficiency of skill acquisition.

In its early development, AQA technology relied heavily on handcrafted feature extraction, which required the design of features that could describe action quality based on expert knowledge and experience. These handcrafted features were often based on kinematic properties such as joint angles, speed, and posture changes. For example, Huang et al. [14] used Kinect sensors to detect joint positions and evaluate patient movements for rehabilitation feedback; Gharasuie et al. [15] assessed movement correctness by quantifying features such as range, duration and speed, offering exercise guidance. However, these methods had limitations. The design of handcrafted features depended on domain experts, making them prone to subjectivity and human bias. This dependency also increased the complexity and cost of feature extraction. Handcrafted features were often tailored to specific tasks or actions, lacking generalization and adaptability in different scenarios. Differences in action quality are often reflected in fine details at the micro level, which handcrafted features struggle to capture effectively. These limitations hindered the performance of handcrafted features in complex and diverse action evaluation tasks.

In recent years, numerous researchers have focused on the use of computer vision and deep learning techniques to address the AQA task. Powered by large datasets, deep learning models can automatically learn and extract complex spatiotemporal features, enabling them to capture subtle differences in actions. For example, Parmar et al. [7] employed 3D convolutional networks (3D ConvNets) to extract features from action videos, followed by a regression model for scoring prediction. This approach significantly outperformed traditional methods based on handcrafted features. Li et al. [16] proposed an end-to-end framework for automatic scoring in sports. This framework utilizes 3D ConvNets to extract features from action segments, enabling simultaneous feature extraction in temporal and spatial dimensions to capture richer and more complex dynamic information. Bai et al. [17] extended the transformer architecture and introduced a Temporal Parsing Transformer (TPT) model specifically designed for AQA tasks. The introduction of these deep learning techniques has provided new approaches for the implementation of AQA tasks.

In this review, we examine the AQA methods developed up to 2025, most of which rely primarily on visual information. In the visual modality, data typically take the form of RGB videos, skeletal data, optical flow data, among others. RGB videos, being the most common visual data type, provide rich color and motion information. Skeletal data extract joint location information, which helps in analyzing the spatial relationships of movements. Optical flow data utilize pixel motion information in image sequences, which is useful for capturing dynamic changes in the temporal dimension. Additionally, in recent years, multi-modal methods have gained widespread attention. These techniques combine different types of data to take advantage of the strengths of each, thereby enhancing the performance of the model. Therefore, we will also review AQA methods based on multi-modal techniques. Otherwise, we focus on AQA methods that are more oriented toward real-world applications.

Classification of literature

Figure 4 illustrates the distribution of AQA-related publications collected between 2014 and 2025 based on our retrieval strategy. Each bar in the figure consists of two lines: the first indicates the year, and the second represents the number of publications for that year. The data are arranged in ascending order by year from left to right. Overall, the number of publications exhibits a fluctuating upward trend. From 2014 to 2018, the annual publication volume remained relatively low. Since 2019, a marked increase can be observed, with the number of publications reaching 35 in 2024. Notably, this high level of research activity is sustained in 2025, with 32 publications, indicating continued and strong interest in AQA research in recent years.

To facilitate a deeper understanding of the similarities and differences between various research directions and methods, this review systematically categorizes the collected literature. In the classification process, we applied different strategies to divide the literature into distinct categories, with the aim of providing a clearer overview of the current research landscape and trends in AQA. Specifically, the literature is organized according to two main classification methods: application-based classification and modality-based classification.

Figure 5 shows the hierarchical classification structure we used to categorize the AQA literature. To systematically organize AQA-related research, we employed two distinct classification strategies: application-based and modality-based classification. In the application-based classification framework, we categorized AQA methods into three main areas: sports, healthcare, and skill assessment, based on their primary application scenarios. In the modality-based classification framework, we divided the AQA methods into single-modal and multi-modal fusion approaches, depending on the number of data modalities utilized. Furthermore, Fig. 5 also lists some relevant literature within each category.

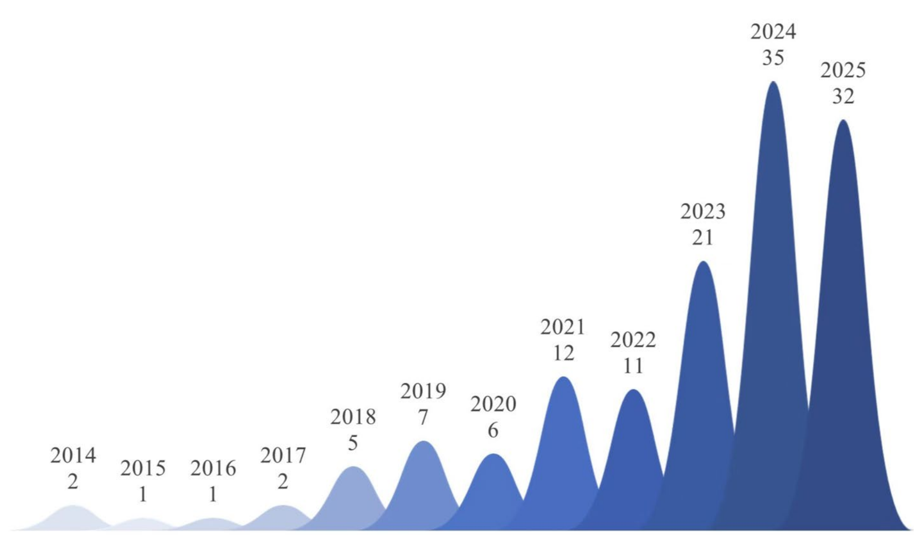

Fig. 4 Yearly trends of AQA literature

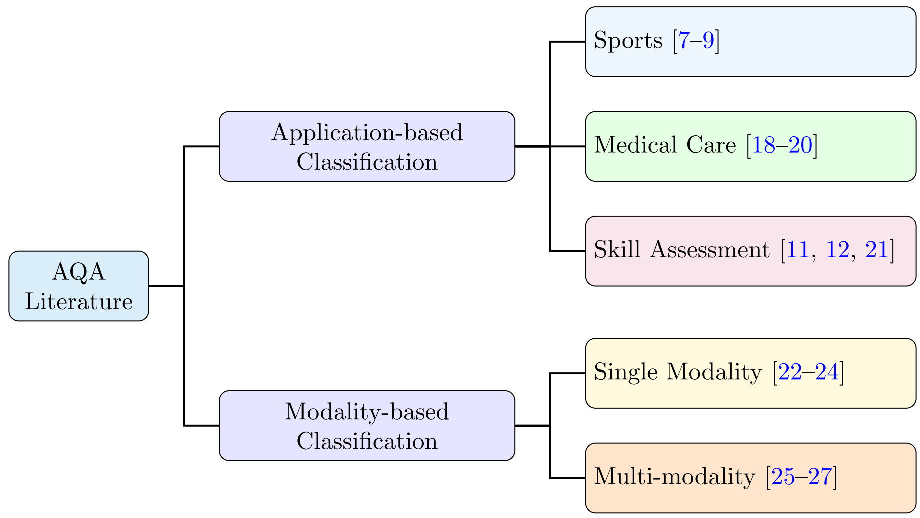

Fig. 5 Structure of AQA literature classification

Classification based on application areas

Classification based on application areas aims to categorize the literature into subfields according to the specific needs of different domains and tasks. Through an analysis of the collected literature, we identified three primary application areas for AQA tasks: sports, medical care, and skill assessment.

In the sports field, AQA tasks are widely applied in athlete performance analysis and competition scoring [7–9, 25, 28, 29]. By employing AQA models, the evaluation of athletes' movements and techniques can be automated, providing precise feedback for coaches and athletes. This helps athletes improve training efficiency and refine their technical skills. The use of AQA models also reduces subjective factors in human scoring, enhancing the objectivity and consistency of evaluation results.

In the medical field, AQA tasks are primarily applied to physical rehabilitation for patients [18–20, 30]. It should be noted that the evaluation of surgical performance is categorized under the skill assessment field. The core of surgical performance evaluation lies in assessing the surgeon's skill level during procedures, including precision and coherence of steps. Since it shares similar characteristics with skill analysis, it is classified under the skill assessment domain. When applied to physical rehabilitation, AQA models can provide real-time feedback to clinicians, helping them develop appropriate quality plans.

In the skill assessment domain, AQA tasks are primarily used to objectively quantify the performance of professionals. This area encompasses various technical tasks, including surgical performance assessment and daily living skills [11, 12, 21, 31]. In these scenarios, AQA models provide detailed analyses of individuals' skills, supporting their skill improvement efforts.

Classification based on modalities

Many AQA methods are primarily based on a single modality, such as using only RGB video or skeleton data for evaluation [7, 23]. However, as the application areas of AQA continue to expand, the limitations of single-modality methods become increasingly evident in complex scenarios. For example, RGB video is susceptible to environmental lighting changes, occlusions, and other factors, and during the evaluation process, the background may also interfere with the results. In multiperson scenarios, skeleton data may suffer from occlusions of certain joints, leading to data loss or inaccuracies. To overcome these limitations, many researchers have begun in recent years to explore multi-modal AQA methods, which combine the strengths of multiple data sources to address the shortcomings of single-modal approache.

We categorized the collected literature into single-modality methods and multi-modal methods. Single-modality methods typically rely on a single type of data to perform the AQA task, with common data types including RGB video or skeleton data [22–24]. Multi-modal methods, on the other hand, combine data from various modalities, such as RGB video, skeleton data, audio, and text, to perform the AQA task [25–27].

Datasets and evaluation metrics

This section provides a systematic overview of datasets and evaluation protocols commonly used in AQA. Existing AQA datasets span diverse application domains, including sports, medical care, and skill assessment, and exhibit substantial variations in annotation granularity, label structure, temporal scope, and available data modalities. These factors play a critical role in shaping model design choices, supervision strategies, and evaluation protocols. We first review representative AQA datasets across different domains, highlighting their key characteristics in terms of provided modalities, annotation content, and dataset scale. To facilitate structured comparison, we summarize different categories of datasets in a series of tables, systematically reporting the available modalities, the number of covered action categories, dataset scale, temporal scope (distinguished as long-term L and short-term S and explicitly indicated in the tables), and label structures. For publicly available datasets, the tables further provide their official access links. Based on these summaries, we analyze the differences among datasets from a taxonomy-driven perspective, aiming to offer practical guidance for dataset selection across different AQA paradigms.

To provide a more intuitive overview, we also present a visual summary in Figure 6, illustrating the modality coverage and application domains of existing AQA datasets.

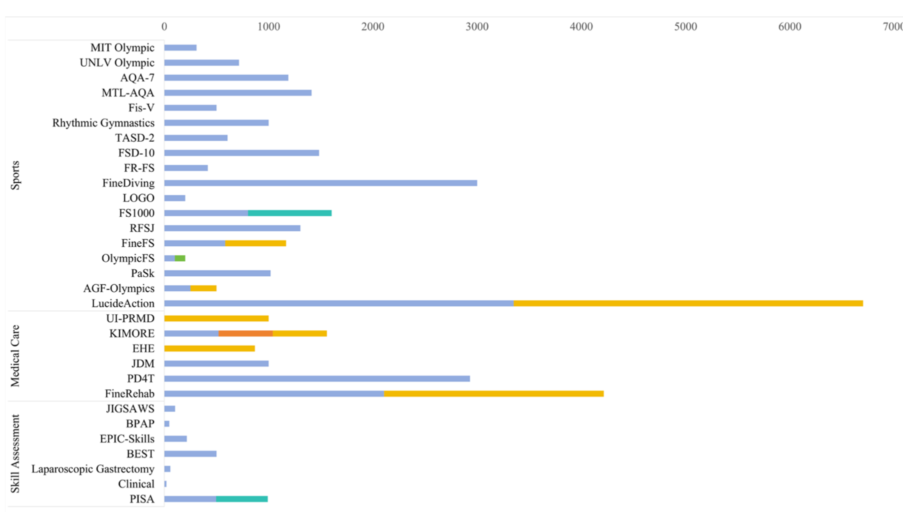

Fig. 6 Modalities and application domains covered by AQA datasets

Each dataset differs significantly in terms of the data modalities it provides, such as RGB, skeleton, audio, and text, as well as the specific types of actions it focuses on. This figure highlights both the diversity and the limitations of current AQA resources, offering insights into trends in modality usage and potential directions for future dataset development.

Finally, we introduce commonly used evaluation metrics for different AQA tasks and discuss their applicability under varying annotation granularities and label structures, enabling a more comprehensive understanding of how AQA models are assessed.

AQA datasets in sports

We reviewed various AQA datasets in the sports domain, mainly originating from Olympic sports such as figure skating, diving, and rhythmic gymnastics. The characteristics, sources and other relevant information for each dataset are presented in detail in Table 1.

We first review AQA datasets that rely exclusively on RGB video data. These datasets constitute the earliest and most widely used benchmarks in AQA research, enabling models to learn performance evaluation directly from visual appearance and motion cues without auxiliary modalities such as pose or audio.

MIT Olympic [9]. The dataset includes data from two sports: diving and figure skating. The MIT diving dataset consists of 159 slow-motion videos, with a frame rate of 60 frames per second, totaling approximately 25,000 frames, and scores ranging from 20 to 100 points. The MIT skating dataset contains 150 videos with a frame rate of 24 frames per second, accumulating around 630,000 frames, with scores ranging from 0 to 100 points. Both of them allocate 100 videos for training, with the remaining videos used for testing. Furthermore, the annotation of the MIT diving dataset was guided by the MIT diving team coach. In contrast, the lower frame rate of the MIT skating dataset results in greater variation in movement, making it more challenging to assess.

UNLV Olympic [7]. This dataset includes three types of sports: diving, figure skating, and vault. The UNLV-Dive dataset is an extension of the MIT-Dive dataset, increasing the original 159 samples to 370, offering a more extensive collection of diving motion data, with 300 samples designated for training and the rest of them used for testing. The UNLV-Skating dataset consists of 170 video samples, with 100 used for training and the remaining 70 for testing. The UNLV-Vault dataset contains 176 vault video samples. Each video averages around 75 frames in length, with score labels based on the sum of execution and difficulty scores.

AQA-7 [32]. The AQA-7 dataset includes samples from seven sports: singles diving-10m platform, gymnastic vault, big air skiing, big air snowboarding, synchronous diving-3m springboard, synchronous diving-10m platform, and trampoline. There are 1,189 video samples in total.

MTL-AQA [33]. The MTL-AQA dataset focuses on diving and includes 1,412 video samples from 16 different events. These samples cover both the 10 m platform and the 3 m springboard events, featuring performances from male and female athletes in individual and pairs of synchronized divers. The dataset provides fine-grained annotations, recording detailed diving movements, difficulty levels of dives, and descriptions of the moves, along with the judges' scores for each dive. Diving actions are further broken

Table 1 Characteristics of AQA datasets in sports

Dataset and Link	Modality	Categories	Samples	Temporal Scope	Annotation Type
MIT Olympic [9]	Video	2	309	S (Dive)/L (Skate)	Score
UNLV Olympic [7] https://rtis.oit.unlv.edu/datasets.html	Video	3	716	S (Dive Vault)/L (Skate)	Score
AQA-7 [32] https://rtis.oit.unlv.edu/datasets.html	Video	7	1189	S (Dive Vault etc.)/L (Trampoline)	Score
MTL-AQA [33] https://github.com/ParitoshParmar/MTL-AQA	Video	16	1412	S	Score
RG [28] https://github.com/qinghuannn/ACTION-NET	Video	1	1000	S	Score
TASD-2 [34] https://www.isee-ai.cn/~gaojibin/ProjectAIM.html	Video	2	606	S	Score
FR-FS [35] https://github.com/Shunli-Wang/TSA-Net	Video	1	417	L	Score
Fis-V [8] https://github.com/chmxu/MS_LSTM	Video	1	500	L	TES, PCS
FSD-10 [36]	Video	10	1484	L	Score, Action
FineDiving [37] https://github.com/xujinglin/FineDiving	Video	1	3000	S	Score, Action, Boundaries
LOGO [38] https://github.com/shiyi-zh0408/LOGO	Video	26	200	L	Score, Action, Boundaries
RFSJ [29]	Video	23	1304	L	Score, Warning Signs, Boundaries
PaSk [39]	Video	1	1018	L	Score, Gain or Error Score
FineFS [40] https://github.com/yanliji/FineFS-dataset	Video, Skeleton	1	1167	L	TES, PCS, Action, Boundaries
AGF-Olympics [25]	Video, Skeleton	1	500	L	Score, Detail Score
LucideAction [41]	Video, Skeleton	8	6702	L	Score, Detail Score
OlympicFS [26]	Video, Text	1	200	L	TES, PCS, Commentary
FS1000 [42]	Video, Skeleton	8	1604	L	TES, PCS

down into the position of the dive, the number of somersaults (SS), and the number of twists (TW).

Fis-V [8]. The Fis-V dataset is a large-scale figure skating video dataset containing 500 videos about ladies' singles short program, with an average duration of 2 minutes and 50 seconds per video. 400 samples are used for training and 100 samples are used for testing. Each video is scored by nine different judges, offering two evaluation criteria. Total Element Score (TES) and Total Program Component Score (PCS).

Rhythmic Gymnastics (RG) [28]. The RG dataset includes videos from four gymnastics events: ball, clubs, hoop, and ribbon. The dataset contains 1,000 videos from the 36th and 37th International Artistic Gymnastics Competitions, with 250 samples for each event. All samples are randomly divided, with 200 samples used for training and 50 samples reserved for testing. Each video has an average length of 1 minute and 35 seconds and a frame rate of 25 frames per second. Three scores are annotated for each video: difficulty score, execution score, and total score.

TASD-2 [34]. The TASD-2 dataset includes two types of synchronized diving actions: 3 m springboard and 10 m platform. The dataset contains 119 samples of 3 m springboard dives and 184 samples of 10 m platform dives, each video consisting of 102 frames. To increase the size of the dataset, data augmentation was applied, resulting in a final total of 238 3 m springboard samples and 368 10 m platform samples. Among the 3 m springboard samples, 188 are used for training and 50 for testing, while 293 of the 10 m platform samples are designated for training and 75 for testing.

FR-FS [35]. The FR-FS dataset includes 417 video clips sourced from the Fis-V dataset and the 2018 Winter Olympic Games in Pingchang, focusing primarily on athletes' take-off, rotation, and landing. The dataset has been randomly partitioned, with 50% of the samples designated for training and the remaining 50% assigned for testing. Of these videos, 276 show athletes landing smoothly, while the remaining 141 capture moments of falls or errors.

FSD-10 [36]. The FSD-10 dataset consists of 1,484 video clips from the 2017 to 2018 Worldwide Figure Skating Championships, covering 10 different movements in both men's and women's events. The videos are recorded at 30 frames per second, with a resolution of 1080×720. The dataset provides fine-grained annotations of actions, with experts meticulously labeling type, grade of execution, skater info, etc.

FineDiving [37]. The FineDiving dataset consists of 3,000 video samples, featuring 52 action types, 29 subaction types, and 23 difficulty levels. The dataset provides fine-grained labels, including action types, subaction types, coarse-grained and fine-grained temporal boundaries, and action scores. Of the total dataset, 75% of the samples are allocated for training, with the remaining 25% reserved for testing.

LOGO [38]. The LOGO dataset is the first fine-grained video dataset designed for group AQA tasks, comprising 200 video samples across 26 events. Each video has an average duration of 204.2 seconds, with a total duration exceeding 11 hours. The dataset includes 3 annotation types, 12 action types, and 17 formation types. Of the 200 samples, 150 are allocated for training, while 50 are used for testing.

RFSJ [29]. The RFSJ dataset is the first AQA dataset that includes replay information. It contains 768 live video sequences and 536 replay sequences. This dataset covers 10 single action types and 13 combination action types. Each video is meticulously annotated with details such as the start and end frame indices of every jump, the correspondence between replay and live sequences, base scores for each action, execution scores from judges, and the final score. In addition, the dataset includes specific warning signs unique to figure skating competitions, identifying actions that do not meet the requirements.

PaSk [39]. The dataset contains 1018 samples, including pair free-skating and pair short programs. Each video is annotated with the final score, as well as the gain or error score, a sub-score used to adjust the total action score. The gain score is a positive decimal, while the error score is a negative decimal. All samples are divided into a training set and a testing set in a 4:1 ratio.

From a methodological perspective, the core difference in AQA datasets that only provide the video modality lies in the granularity of annotations, which directly influences the choice of modeling strategies. Datasets that provide only overall or component-level scores, such as MIT Olympic [9], UNLV Olympic [7], AQA-7 [32], and Fis-V [8], are well suited for global score regression approaches. The lack of fine-grained supervision in these datasets also motivates weakly supervised and attention-based methods that implicitly identify informative frames or phases. In contrast, datasets with richer annotations-such as FSD-10 [36], FineDiving [37], and PaSk [39] are better support fine-grained AQA methods that explicitly model action structure and temporal dynamics. The availability of action-level labels, temporal boundaries, synchronization cues, or error descriptions enables research on phase-aware modeling, temporal localization, multi-task learning, and interpretable assessment, facilitating the evolution of AQA from global score prediction toward structured and explainable quality analysis.

Beyond RGB videos, several datasets incorporate human pose information to explicitly model body dynamics and motion structure, which is particularly beneficial for fine-grained quality assessment. From a multi-modal perspective, pose representations provide a complementary, geometry-aware modality that is largely invariant to appearance, illumination, and background clutter, thereby facilitating more robust and interpretable multi-modal fusion for AQA.

FineFS [40]. The FineFS dataset contains 1,167 figure skating video samples, with 729 samples from short programs, each with an approximate duration of 2 minutes and 40 seconds, and 438 samples from free skating programs, averaging around 4 minutes. All videos have a frame rate of 25 frames per second. The dataset offers fine-grained annotations, including four levels of score labeling: total score (TOT), TES and PCS, etc. It also provides two levels of action categorization: coarse-grained and fine-grained. In addition, time annotations are provided for each action segment. Of the total samples, 933 are used for training and 234 are used for testing.

AGF-Olympics [25]. The AGF-Olympics dataset provides two modalities: RGB videos and 2D joint position information (skeleton representation). It contains 500 artistic gymnastic floor routines video samples, with a resolution of 256x256. The videos range in duration from 1 minute and 18 seconds to 2 minutes, with an average of 2,500 frames per video. OpenPose was used to extract the 2D skeleton joints from the videos. The dataset includes three types of annotations: difficulty score, execution score, and penalty score.

LucideAction [41]. The dataset includes eight different sports events, such as floor exercise, vault, and parallel bars, comprising a total of 6,702 samples. Each sample provides multi-modal data, including RGB videos and 2D and 3D pose sequences. Every action sample is annotated with performance scores, along with detailed breakdowns of errors and corresponding deductions in various aspects of the movement execution.

These datasets all provide rich fine-grained annotations and explicit pose representations, making them particularly well suited for fine-grained and structure-aware AQA methods. The availability of 2D and 3D skeleton information enables models to explicitly capture body kinematics, motion coordination, and geometric consistency, which facilitates pose-guided temporal modeling and cross-modal fusion.

Beyond visual appearance and pose information, some AQA datasets further incorporate additional modalities such as textual descriptions or audio signals to capture complementary semantic and contextual cues. These modalities provide higher-level information that is difficult to infer from visual data alone, such as rhythm, musicality, and expert judgment, thereby enabling more comprehensive multi-modal AQA.

OlympicFS [26]. This dataset includes 200 high-quality figure skating videos from professional international competitions, including the 2018 PyeongChang Winter Olympics and the 2022 Beijing Winter Olympics, with a resolution of 1280×720 pixels. Of these, 160 samples are used for training and 40 for testing. The dataset is richly annotated, including PCS, TES, program categories, and expert commentary for each video sample. This dataset supports multi-modal AQA tasks.

FS1000 [42]. The FS1000 dataset consists of 1,604 figure skating videos, each video containing approximately 5,000 frames at a frame rate of 25 frames per second. Of all samples, 1,247 are assigned for training. The videos vary in length, ranging from 2.5 to 4.3 minutes, with an average duration of around 3 minutes and 20 seconds. Each video is annotated with official scores, which include the TES and the PCS.

By incorporating non-visual modalities such as textual commentary and audio information, these datasets are well suited for semantic-aware and cross-modal AQA methods. They enable multi-modal fusion frameworks that combine visual features with high-level contextual cues, such as cross-modal attention and representation alignment. Moreover, the availability of expert commentary supports interpretable and explanation-oriented AQA approaches, where textual cues can guide score prediction or provide human-readable justifications for model decisions, extending AQA beyond purely motion-driven evaluation.

AQA datasets in medical care

In this section, we review AQA datasets in the medical care domain. Compared to the sports domain, medical care datasets are more difficult to collect due to factors such as patient privacy. As a result, there are relatively fewer available datasets in this domain, and publicly accessible datasets are especially rare. These datasets primarily focus on rehabilitation treatments, and the details of each dataset are presented in Table 2.

Beyond sports scenarios, AQA has also been explored in the medical and healthcare domain, where the goal is to quantitatively evaluate movement quality for disease assessment, rehabilitation monitoring, and functional analysis. Medical AQA datasets typically focus on patients or elderly individuals performing predefined movements or daily exercises, and the evaluation criteria are closely related to clinical scales, motor function grades, or abnormality detection. existing medical AQA datasets rely on either RGB video or skeleton data.

Table 2 Characteristics of AQA datasets in medical care

Dataset and Link	Modality	Categories	Samples	Temporal Scope	Annotation Type
PD4T [20] https://github.com/Plrbear/PECoP	Video	4	2931	S	Score
JDM [43]	Video	14	over 1000	S	Grade
UI-PRMD [18] https://www.uidaho.edu/404.htm	Skeleton	10	1000	S	Category
EHE [44]	Skeleton	6	869	S	Category
FineRehab [30] https://bsu3dvlab.github.io/FineRehab/	Video, Skeleton	15	4215	S	Score
KIMORE [19]	Video, Depth, Skeleton	5	1560	S	Score

PD4T [20]. The PD4T dataset contains 2,931 video samples of 30 patients with Parkinson's Disease (PD) tested at different time intervals. The original videos have a resolution of 1920x1080, which was adjusted to 854x480, with a frame rate of 25 frames per second. These videos are scored by clinicians using the Unified Parkinson's Disease Rating Scale (UPDRS), with scores ranging from 0 (normal) to 4 (severe), to assess the quality of movement and severity of symptoms in patients.

JDM [43]. The dataset contains samples of various actions performed by children with juvenile dermatomyositis (JDM), with a total of more than a thousand samples. It includes 14 distinct movement categories. Each action sample is associated with a corresponding grade label, indicating the level of muscle strength for that particular action.

UI-PRMD [18]. The UI-PRMD dataset comprises motion data from 10 healthy participants, each performing 10 different physical therapy exercises, each exercise repeated 10 times, resulting in a total of 1,000 motion samples. These data were simultaneously collected using the Vicon optical tracking system and the Microsoft Kinect sensor, capturing both joint angle and positional information.

EHE [44]. The dataset includes 25 elderly people performing six types of morning exercises in a nursing home, with a total of 869 action repetitions. Ten of the elderly people were diagnosed with varying degrees of Alzheimer's Disease (AD). The data type is skeleton data. Actions deemed abnormal are labeled as"abnormal,"while the remaining normal actions are labeled as"normal."

From a task and methodology perspective, these medical AQA datasets support different problem formulations. PD4T [20] provides continuous clinical scores and is therefore well suited for regression-based AQA methods that aim to predict movement severity or symptom progression. In contrast, JDM [43], UI-PRMD [18], and EHE [44] provide discrete grade or category labels, making them more suitable for classification-based approaches, such as movement quality grading, or abnormality detection. Given the relatively small number of subjects and strong clinical constraints, these datasets particularly favor lightweight temporal modeling, skeleton-based analysis, and interpretable learning frameworks that can support reliable and explainable medical decision-making.

Recent research has increasingly explored multi-modal sensing to capture complementary information for rehabilitation assessment. By jointly leveraging RGB videos, depth maps, and skeletal motion, these datasets provide a more comprehensive description of human movement, enabling robust evaluation of execution quality, coordination, and smoothness.

FineRehab [30]. The FineRehab dataset includes data from 50 participants who performed 16 different rehabilitation exercises, consisting of both individuals with musculoskeletal disorders and healthy participants, with a total of 4,215 movement samples. The data were captured using two Kinect cameras and 17 IMU sensors, covering both RGB video data and skeletal data modalities. 75% of the data is used for training, while 25% is reserved for testing. The dataset evaluates the 16 specific rehabilitation exercises in three dimensions: completeness, correction, and smoothness, with scores ranging from 0 to 4. Finally, an overall quality score for each exercise is calculated based on the weights assigned by experts to different parts of the body and the dimensions of the evaluation.

KIMORE [19]. The KIMORE dataset was collected using RGB-D sensors to capture data from healthy individuals and patients with movement disorders performing five common lower back rehabilitation exercises. The data collected includes RGB video, depth video, and skeletal joint position and orientation information. In addition, the dataset provides clinical features for each movement along with the corresponding clinical scores.

By integrating multiple sensing modalities such as RGB video, depth information, and skeletal motion, these datasets are well suited for multi-modal and structure-aware medical AQA methods. They naturally support fusion-based approaches that combine complementary motion cues across modalities, as well as multi-task learning frameworks that jointly model overall quality scores and dimension-specific evaluations. Moreover, the availability of clinically meaningful scores and expert-defined evaluation criteria makes these datasets particularly suitable for interpretable and clinically grounded AQA models, which aim to support reliable rehabilitation assessment and personalized therapy feedback.

AQA datasets in skill assessment

AQA datasets in the skill assessment domain cover a wide range of human skills, from simple everyday tasks to more complex skills such as surgical procedures. These datasets provide valuable data for assessing the execution and proficiency of various skills. The basic characteristics of each dataset are summarized in Table 3.

Skill assessment datasets typically rely on RGB videos, with some datasets incorporating additional modalities such as audio. Annotations vary from absolute scores and

Table 3 Characteristics of AQA datasets in skill assessment

Dataset and Link	Modality	Categories	Samples	Temporal Scope	Annotation Type
EPIC-Skills [11]	Video	4	216	L	Pairwise Ranking
BEST [12] https://github.com/hazeld/rank-aware-attention-network	Video	5	500	L	Pairwise Ranking
BPAP [45]	Video	1	48	L	Event
Laparoscopic Gastrectomy [46]	Video	1	57	L	Score
JIGSAWS [21] https://cirl.lcsr.jhu.edu/research/hmm/datasets/jigsaws_release/	Video	3	103	L	Score, Gesture Label
Clinical [31]	Video	1	20	L	Score, Event
PISA [13]	Video, Audio	1	992	L	Grade, Bounding Box

categorical skill levels to pairwise rankings and fine-grained event or gesture labels, enabling a wide range of learning paradigms for skill-oriented AQA.

EPIC-Skills [11]. This dataset includes four types of skill tasks: surgery, dough-rolling, drawing, and chopstick using, with a total of 216 video samples. Surgery video samples are sourced from the JIGSAWS dataset, and the annotations consist of expert-provided scores. The remaining three tasks are annotated using pairwise comparisons.

BEST [12]. The BEST dataset covers five types of everyday skill tasks: scrambling eggs, braiding hair, tying a tie, making an origami crane, and applying eyeliner. Each task includes 100 video samples, with the start and end points of the actions annotated in every video. The dataset classifies the samples into three skill levels: Beginner, Intermediate and Expert. In addition, 40% of the video samples have been annotated with pairwise comparisons.

BPAP [45]. This dataset features basketball videos shot from a first-person perspective and includes 48 video samples. Each video averages approximately 13 minutes in length, with a total dataset duration of 10.3 hours, capturing real game scenarios of 48 college basketball players. The videos are recorded at a resolution of 1280×960 and a frame rate of 100 frames per second. Of these, 24 videos are used for training. The dataset also includes detailed annotations for three key basketball events: someone shooting a ball, the camera wearer possessing the ball, and a shot made. These events are annotated with 3,734, 4,502, and 2,175 labels, respectively, providing a wealth of annotated data for researchers to perform precise analysis and evaluation of critical actions in basketball games.

Laparoscopic Gastrectomy [46]. The dataset consists of 57 laparoscopic gastrectomy videos, specifically for the treatment of gastric cancer. These videos were recorded using built-in cameras with a resolution of 960x540 and a frame rate of 25 frames per second. The dataset focuses on the infrapyloric area of gastrectomy, with skill assessments conducted for this area. The surgeries range in duration from 8 to 57 minutes, with an average duration of approximately 26 minutes. Each surgical video is annotated by six surgeons using 14 different metrics, with the average score from the senior surgeons serving as the ground truth for these metrics.

JIGSAWS [21]. The JIGSAWS dataset includes three fundamental surgical tasks: suturing, knot tying, and needle passing. There are 39 video samples for suturing, 36 for knot tying, and 28 for needle passing. All videos have a frame rate of 30 frames per second and a resolution of 640x480. The dataset includes detailed gesture labels and scores for each surgical task.

Clinical [31]. This clinical dataset consists of 20 laparoscopic surgery videos focused on the treatment of gastric cancer, including partial or total gastrectomy and related lymph node dissection. Each video has a resolution of 960x540, a frame rate of 25 frames per second, and an average duration of 199 minutes, which makes them relatively long. The dataset provides fine-grained annotations, including skill assessments based on OSATS standards, a proxy score of field clearness, and detailed labeling of 41 types of comprehensive surgical event.

PISA [13]. The dataset includes 61 piano performance videos. The videos are sampled every 160 frames, resulting in 992 independent samples. Of these, 516 samples are used for training and 476 samples for testing. Each sample is annotated with the player's skill level, song difficulty level, name of the song, and a bounding box around the pianist's hands.

Datasets with pairwise comparison annotations, such as EPIC-Skills [11] and BEST [12], are particularly suitable for ranking-based and relative skill learning methods that focus on comparative judgment rather than absolute score prediction. Datasets providing continuous scores or expert-defined metrics, including Laparoscopic Gastrectomy [46], JIGSAWS [21], and Clinical [31], naturally support regression-based approaches and multi-task frameworks that jointly model overall skill level and fine-grained procedural events or gestures. Event-annotated datasets such as BPAP [45] facilitate temporal localization and event-aware skill assessment, while datasets like PISA [13], which incorporate audio cues and hand-level annotations, are well suited for multi-modal and region-focused AQA methods. Overall, these datasets play a crucial role in advancing skill assessment from coarse proficiency estimation toward fine-grained, interpretable, and task-aware quality evaluation.

Evaluation metrics

AQA tasks can generally be categorized into three types based on the nature of the evaluation output: regression, ranking, and pairwise ranking. Regression tasks aim to predict continuous scores that closely approximate human annotations, making them suitable for scenarios where objective scoring criteria are available. Ranking tasks, on the other hand, focus on producing an overall order of multiple samples according to their relative quality, emphasizing global consistency. Pairwise ranking tasks further narrow the scope to learning the relative quality between pairs of samples and modeling fine-grained comparisons. Due to differences in task objectives and output formats, each type of AQA task typically adopts specific evaluation metrics that best capture the intended assessment goals. In the following, we introduce the commonly used evaluation metrics for each task type to facilitate a clearer understanding of how AQA model performance is measured.

Regression task

The goal of regression tasks is to assign a score to the target action, reflecting its level of quality. For regression tasks, a commonly used metric is the Spearman's Rank Correlation Coefficient (SRCC). SRCC measures the monotonic relationship between two variables, with values ranging from −1 to 1. A higher SRCC value, closer to 1, indicates better model performance. The formula for calculating SRCC is as follows:

$$\rho = \frac{\sum_i (p_i - \mu_p)(q_i - \mu_q)}{\sqrt{\sum_i (p_i - \mu_p)^2 \sum_i (q_i - \mu_q)^2}}, \tag{1}$$

where p_i and q_i represent the ranking sequences of the predicted and ground truth scores, respectively. μ_p and μ_q represent the average values of these two sequences.

Yu et al. [22] introduces a more stringent evaluation metric, called the relative L2-distance($\mathrm{R}\text{-}\ell_2$), specifically designed for AQA regression tasks. By accounting for intra-class variance, it provides a more precise measure of the performance in AQA. The computational formula is presented as follows:

$$\text{R-}\ell_2(\theta) = \frac{1}{N}\sum_{i=1}^{N}\left(\frac{|s_i - \tilde{s}_i|}{s_{max} - s_{min}}\right)^2, \tag{2}$$

where s_i and $\tilde{s}_i$ represent ground truth and predicted scores for i-th sample, respectively. s_{max} and s_{min} represent the maximum and minimum scores that can be achieved for an action.

Ranking task

In ranking tasks, action quality is divided into several discrete levels, with each level representing a different quality, such as "excellent,""average"and"poor."The task typically uses accuracy as an evaluation metric and its calculation formula is as follows:

$$\text{Accuracy} = \frac{p}{q} \times 100\%, \tag{3}$$

where p is the number of correctly classified samples and q is the total number of samples.

Pairwise sorting task

For pairwise sorting tasks, the objective of the model is to compare two sets of actions and determine which set exhibits higher quality. A commonly used evaluation metric is pairwise accuracy and its calculation formula is as follows:

$$\text{Acc}_{\text{pair}} = \frac{m}{n} \times 100\%, \tag{4}$$

where m is the number of pairs ordered correctly and n is the total number of video pairs.

Review of existing AQA methods

In this section, we present a systematic and comprehensive review of existing AQA methods, covering both early handcrafted-feature-based approaches and recent advances driven by deep learning. The discussion primarily focuses on deep learning–based AQA methods, with an in-depth analysis of their implementation pipelines and key innovations in feature extraction, network architecture design, and loss function formulation. Our goal is to provide readers with a structured understanding of the current research landscape as well as the underlying methodological motivations that have shaped the field.

We first review traditional AQA methods (Sect. "Traditional methods"), which mainly rely on hand-crafted features and classical regressors to establish the early foundations of the task. We then move to deep learning-based AQA (Sect. "General deep learning methods") following an increasingly fine-grained modeling pipeline: starting from general deep frameworks with regression heads, and further advancing to fine-grained temporal structure parsing and segmentation-based modeling, interaction-aware and structured motion reasoning, and uncertainty-aware score estimation; we also separately summarize attention-based modeling as an important mechanism widely used to enhance representation learning and temporal reasoning. Next, we extend the discussion from single-video prediction to pairwise learning paradigms (Sect. "Contrastive learning based methods"). Building upon fully supervised settings, we further cover learning and generalization regimes including semi-supervised learning, domain shift adaptation, continual learning, and causality- and rule-aware AQA (Sects. "Semi-supervised learning based methods"–"Causality-based and rule-aware methods"). Finally, we review multimodal AQA (Sect. "Multi-modal based methods"), which integrates complementary cues beyond RGB (e.g., pose, audio, or semantics) to improve robustness and interpretability. This organization highlights how AQA methods evolve from feature engineering to structured and robust deep models, and from standard supervision to broader real-world learning scenarios.

Accordingly, this section organizes existing AQA methods following this methodological evolution, and systematically reviews representative approaches ranging from traditional and general deep learning methods to contrastive, semi-supervised, domain shift–aware, continual learning, causality-based, and multi-modal frameworks. To further contextualize the development of the field, Fig. 7 presents a timeline of representative AQA methods from 2014 to 2025, illustrating the evolution from handcrafted feature extraction to deep learning, and more recently to multi-modal, semi-supervised, and causality-aware approaches. This timeline highlights key milestones and emerging trends, offering a concise overview of the past decade of AQA research.

Traditional methods

Early approaches to AQA primarily relied on handcrafted features combined with conventional regression or classification models. These methods generally follow a two-stage paradigm: first designing task-specific spatio-temporal descriptors to characterize motion execution, and then learning a mapping from these features to expert-provided quality scores. Such methods laid the foundation for automated AQA and provided insights into which motion cues are relevant for assessing action quality.

A common strategy involves extracting spatio-temporal motion patterns from video sequences. This typically includes detecting interest points, encoding local motion using descriptors such as HOG and HOF, and modeling temporal variations to capture consistent action dynamics. Regression models are then applied to map these features to quality scores, emphasizing the importance of motion dynamics and temporal consistency in skill assessment [54].

Another prominent approach leverages human pose or skeleton representations. Pose-driven methods extract joint positions or skeletal structures to model spatial and temporal motion characteristics. Techniques such as manifold learning or skeleton alignment

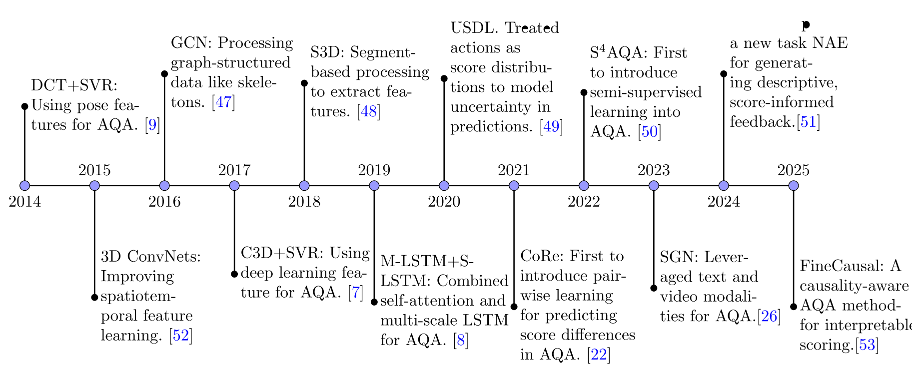

Fig. 7 Timeline of methodological evolution in AQA

quantify deviations from reference motions, enabling interpretable evaluation of movement quality [55, 56]. Spatio-temporal transformations, such as frequency-domain analysis of joint trajectories, further enhance robustness by focusing on low-frequency motion patterns while reducing the influence of frame-level noise [9].

Despite their effectiveness in early applications, traditional methods are constrained by their reliance on handcrafted features and task-specific heuristics. Such features often struggle to capture complex motion patterns and generalize across different actions or domains. Moreover, the separation between feature extraction and score prediction limits the capacity for learning discriminative representations in an end-to-end manner. These challenges have motivated the development of deep learning-based AQA methods, which aim to automatically learn robust spatio-temporal representations directly from data.

General deep learning methods

General deep learning–based AQA methods follow a unified paradigm of learning representations directly from data under supervised settings, while differing in how action quality is modeled and inferred. To provide a clear organization, we categorize these methods according to their primary modeling focus, progressing from holistic representations to more fine-grained, structured, and uncertainty-aware formulations. We first review video-level representation learning and direct regression methods (Sect. "Video-level representation learning and direct regression"), which map an entire video to a scalar quality score. Attention mechanism–based methods (Sect. "Attention mechanism based methods") then improve this formulation by selectively emphasizing informative frames or clips. Temporal structure modeling and segmentation-based methods (Sec. "Temporal structure modeling and segmentation-based methods") further exploit the inherent phase structure of actions, while interaction-aware and structured motion modeling (Sect. "Interaction-aware and structured motion modeling") focuses on explicit relationships among body parts or motion components. Finally, uncertainty-aware and distribution-based score modeling (Sect. "Uncertainty-aware and distribution-based score modeling") addresses annotation ambiguity by modeling score distributions rather than deterministic predictions.

Following this taxonomy, representative general deep learning–based AQA methods are reviewed in this section, with their main characteristics and results summarized in Table 4.

Video-level representation learning and direct regression

Early deep learning approaches to AQA predominantly adopt a video-level, single-stage regression paradigm, in which an entire video is encoded into a holistic spatiotemporal representation and directly mapped to a scalar quality score. These methods are based on the assumption that global visual appearance and motion dynamics are sufficient to characterize overall execution quality.

Within this paradigm, most methods focus on learning effective video-level representations using deep spatiotemporal encoders. Typically, 3D convolutional networks are employed to jointly model appearance and motion cues, followed by temporal aggregation mechanisms that compress frame-level features into a single global embedding. Regression or sequential models are then applied to predict quality scores from these

Table 4 Results of General Deep Learning Methods

Dataset	Method	Publisher	SRCC	R-l_2	Acc_{pair}
MIT-Diving	C3D+SVR [7]	CVPR 2017	0.74	-	-
MIT-Skating	C3D+SVR [7]	CVPR 2017	0.53	-	-
	MSE+Ranking loss [16]	PCM 2018	0.5753	-	-
	C3D+S-LSTM+M-LSTM [8]	TCSVT 2019	0.59	-	-
	SCN+ATCN [57]	PRCV 2020	0.71	-	-
	ACTION-NET [28]	ACM MM 2020	0.615	-	-
UNLV-Diving	C3D+SVR [7]	CVPR 2017	0.78	-	-
	MSE+Ranking loss [16]	PCM 2018	0.8009	-	-
	S3D [48]	ICIP 2018	0.86	-	-
	C3D-AVG-MTL [33]	CVPR 2019	0.8808	-	-
	SCN+ATCN [57]	PRCV 2020	0.85	-	-
	Dong et al. [58]	KBS 2021	0.8798	-	-
	PSL [59]	APIN 2023	0.87	-	-
UNLV-Vault	C3D+SVR [7]	CVPR 2017	0.66	-	-
	MSE+Ranking loss [16]	PCM 2018	0.7028	-	-
	SCN+ATCN [57]	PRCV 2020	0.76	-	-
PaSk	Gao et al. [39]	IJCV 2023	0.64	-	-
TASD-2	AIM [34]	ECCV 2020	0.89	-	-
	Gao et al. [39]	IJCV 2023	0.915	-	-
	GOAT [38]	CVPR 2023	0.9334	0.7054	-
Fis-V	C3D+S-LSTM+M-LSTM [8]	TCSVT 2019	0.725	-	-
	GDLT [60]	CVPR 2022	0.761	-	-
	LUSD-Net [40]	ACM MM 2023	0.76	-	-
MTL-AQA	C3D-AVG-MTL [33]	CVPR 2019	0.9044	-	-
	USDL [49]	CVPR 2020	0.9273	-	-
	Roditakis et al. [61]	PETRA 2021	0.77	-	-
	C3D-AVG-SAHMreg [62]	ICIP 2021	0.897	-	-
	UD-AQA [63]	ARXIV 2022	0.9545	0.259	-
	Zhou et al. [64]	TCSVT 2023	0.9563	0.235	-
	TECN [65]	CAIS 2023	0.9095	-	-
	DAE [66]	NCAA 2024	0.9589	-	-
	FSPN [67]	TIP 2023	0.9601	0.221	-
AQA-7	Pan et al. [68]	ICCV 2019	0.7849	-	-
	USDL [49]	CVPR 2020	0.8192	-	-
	TSA-Net [35]	ACM MM 2021	0.8476	-	-
	Pan et al. [69]	TPAMI 2021	0.85	-	-
	Zhou et al. [64]	TCSVT 2023	0.8501	1.98	-
	TECN [65]	CAIS 2023	0.8504	-	-
	AdaST [70]	TMM 2023	0.8443	-	-
	FSPN [67]	TIP 2023	0.8724	1.56	-
	GOAT [38]	CVPR 2023	0.9325	0.6423	-
	DAE [66]	NCAA 2024	0.852	-	-
FineDiving	FSPN [67]	TIP 2023	0.942	0.278	-
	GOAT [38]	CVPR 2023	0.9032	0.3529	-
RG	ACTION-NET [28]	ACM MM 2020	0.617	-	-
	GDLT [60]	CVPR 2022	0.765	-	-
LOGO	GOAT [38]	CVPR 2023	0.5599	4.7626	-
FineFS	LUSD-Net [40]	ACM MM 2023	0.792	-	-

Table 4 (continued)

Dataset	Method	Publisher	SRCC	R-l_2	Acc_{pair}
JIGSAWS	Pan et al. [68]	ICCV 2019	0.57	-	-
	AIM [34]	ECCV 2020	0.71	-	-
	Pan et al. [69]	TPAMI 2021	0.7815	-	-
	VTPE [31]	CVPR 2021	0.8	-	-
	ViSA [71]	MICCAI 2022	0.83	-	-
	UD-AQA [63]	ARXIV 2022	0.89	4.33	-
	Gao et al. [39]	IJCV 2023	0.88	-	-
	Zhou et al. [64]	TCSVT 2023	0.9	4.031	-
	DAE [66]	NCAA 2024	0.86	-	-
clinical	VTPE [31]	CVPR 2021	0.565	-	-
EPIC-Skills	Siamese TSN [11]	CVPR 2018	-	-	0.76
	Doughty et al. [12]	CVPR 2019	-	-	0.803
	Pan et al. [69]	TPAMI 2021	-	-	0.8406
	AdaST [70]	TMM 2023	-	-	0.8245
BEST	Doughty et al. [12]	CVPR 2019	-	-	0.812
	Pan et al. [69]	TPAMI 2021	-	-	0.8327
	AdaST [70]	TMM 2023	-	-	0.8534
UI-PRMD	EGCN [23]	IJCAI 2022	-	-	0.869
KIMORE	EGCN [23]	IJCAI 2022	-	-	0.801
	EGCN++ [72]	TPAMI 2024	0.5218	-	-

representations. To enhance discriminability, several works further introduce task-driven spatiotemporal modeling strategies, including region-focused feature extraction, multi-path temporal architectures, and explicit separation of spatial and temporal streams. In addition, ranking-based objectives are often adopted to model relative skill differences across samples rather than absolute scores [7, 11, 45].

Beyond RGB-based representations, video-level regression frameworks have been extended by incorporating additional modalities and domain-specific priors. Handcrafted cues reflecting task-relevant attributes can be fused with deep visual features to improve robustness in specialized scenarios, while skeleton-based representations modeled using spatial–temporal graph convolutional networks demonstrate that holistic action quality estimation can generalize beyond raw video data by capturing global human motion patterns [44, 46].

In addition to representation design, training strategies play an important role in improving video-level AQA models. Multi-task learning frameworks jointly optimize quality prediction with auxiliary objectives such as action recognition or action description, encouraging shared representations to encode semantically meaningful motion patterns. Complementary regularization strategies, including adversarial learning and masking-based objectives, are employed to suppress background bias and irrelevant visual cues, thereby enhancing robustness under challenging conditions [33, 62].

The objective functions used in video-level regression have also evolved to better align model outputs with human evaluation criteria. Attention-enhanced temporal modeling combined with composite loss functions improves prediction accuracy and correlation with expert scores. Furthermore, distribution-aware formulations model ground-truth scores as probability distributions rather than deterministic values, explicitly accounting for subjectivity and inter-annotator variability in human judgment [57, 65].

Despite these advances, video-level approaches remain fundamentally constrained by their reliance on holistic representations. Temporal dynamics are captured only implicitly through coarse aggregation, which may dilute critical action moments and obscure phase-dependent quality variations. These limitations motivate subsequent research toward more structured and fine-grained temporal modeling paradigms.

Attention mechanism based methods

Inspired by human visual perception, early studies introduced selective attention mechanisms that sequentially focus on salient regions of an image to extract discriminative information, thereby reducing computational cost while enhancing the modeling of key visual cues [73]. Subsequently, attention mechanisms were further generalized into self-attention, leading to the development of the Transformer architecture, which explicitly models global dependencies among input elements and achieved remarkable success in natural language processing tasks [74]. Building on these advances, the Transformer architecture was extended to the vision domain by representing an image as a sequence of patches and applying self-attention for global feature modeling, demonstrating competitive or even superior performance compared to convolutional neural networks in image recognition tasks [75].

Motivated by the strong representational capacity and computational efficiency of attention-based models, recent studies have incorporated various forms of attention mechanisms into AQA to more effectively capture key spatiotemporal cues, suppress irrelevant background information, and improve evaluation accuracy and overall efficiency. This section provides a systematic review of AQA methods based on attention mechanisms.

Early explorations focused on leveraging attention mechanisms to identify key temporal information within long video sequences. For instance, self-attention was introduced into recurrent architectures to highlight critical frames or motion patterns, enabling more effective extraction of discriminative temporal cues from sequential features [8]. Beyond temporal saliency, attention has also been used to suppress irrelevant spatial information. Wang et al. [35] argue that generic pre-trained feature extractors are suboptimal for AQA and propose the Tube Self-Attention (TSA) module, which integrates performer-centric tube modeling with self-attention to filter background noise and emphasize action-relevant regions. In addition, attention has been employed to facilitate contextual interaction between heterogeneous feature streams, such as global video representations and interaction-aware features, enhancing the expressiveness of fused representations [34, 39].

Some studies uses attention to explicitly model the relative importance of video segments for action quality evaluation. Rank-aware and context-aware attention mechanisms have been proposed to focus on segments that are most indicative of skill differences, rather than treating all parts of an action equally. Typical approaches extract spatiotemporal features from paired videos using 3D CNNs and apply attention to capture decisive moments that influence skill ranking or scoring [12, 28]. Some methods further distinguish between dynamic motion cues and static appearance cues by employing separate streams, where attention modules identify and integrate the most informative components from both modalities before regression [28].

With the increasing adoption of Transformer architectures, attention mechanisms have also been extended to capture richer temporal dependencies and fine-grained action structures. Transformer-based models enhance temporal context modeling through self-attention in encoder modules, while decoder-based attention mechanisms interact video features with task-specific prototypes or latent representations to extract grade-related or skill-related features [60]. Beyond coarse temporal modeling, Gedamu et al. [67] introduce fine-grained spatiotemporal parsing to decompose actions into sub-actions, which are then processed by multi-scale spatiotemporal attention modules to capture dependencies across different temporal resolutions, leading to more detailed and structured action representations.

Attention has also been integrated into modular components that can be embedded into existing AQA frameworks to improve performance. For example, group-aware attention mechanisms combine graph convolution and temporal attention to selectively aggregate information from different video segments, enabling more effective fusion of local clip-level features and global temporal context [38]. Such plug-and-play attention modules offer a flexible way to enhance representation learning without redesigning the entire assessment pipeline.

In more specialized scenarios, attention mechanisms have been employed to disentangle multiple evaluation dimensions within a single action. Ji et al. [40] proposed the Localization-assisted Uncertainty Score Disentanglement Network (LUSD-Net) to predict PCS and TES scores in figure skating. LUSD-Net uses a cross-attention mechanism to separately learn two prototype vectors for PCS and TES from the features extracted from the video. This approach decouples the video features into representations specifically for PCS and TES.

Temporal structure modeling and segmentation-based methods

To overcome the limitations of global pooling, subsequent research places increasing emphasis on explicit temporal modeling, motivated by the observation that action quality often varies across different temporal stages. Rather than treating an action as a single indivisible unit, these methods aim to preserve temporal structure by operating on segments, sequences, or semantically meaningful phases.

A representative class of approaches adopts segment-based temporal decomposition. In this paradigm, action videos are uniformly divided into multiple fragments, each of which is independently encoded using 3D ConvNets or their computationally efficient variants. Segment-level features are then aggregated-via concatenation, pooling, or regression-to predict overall action quality, frequently combined with ranking-aware objectives to enforce relative performance constraints across samples. By shifting modeling granularity from the entire video to fragments, these methods alleviate information dilution caused by global pooling and enable more fine-grained temporal analysis [16, 48]. Building upon segment-level representations, another line of work focuses on strengthening long-range temporal dependency modeling over clip or segment sequences. Instead of relying on simple aggregation operators, sequence modeling modules such as self-attentive LSTM architectures and multi-scale temporal convolutional structures are introduced to capture both local motion details and global temporal context. These designs improve the ability to identify temporally critical moments in long and complex action sequences, resulting in more discriminative video-level representations for AQA [8].

Beyond uniform temporal segmentation, several methods incorporate semantic sub-phase modeling that aligns temporal structure with domain-specific execution patterns and scoring criteria. Actions are decomposed into interpretable stages-either predefined or latent-and spatiotemporal features are extracted and regressed at the phase level before being fused into a global evaluation. To further enforce coherence between phase-level predictions and final scores, auxiliary training strategies and label reconstruction mechanisms are employed to explicitly encode scoring consistency constraints [58, 59]. This paradigm enhances both assessment accuracy and interpretability by explicitly linking temporal structure to evaluation standards.

To address limitations arising from rigid segmentation and inter-segment discontinuities, more recent work explores structure-aware temporal aggregation through relational modeling. Hierarchical or graph-based architectures model interactions among temporal segments at multiple levels, refining clip-level representations and aggregating them into coherent action-level descriptors. By explicitly modeling inter-segment relationships, these approaches reduce sensitivity to segmentation ambiguity and better capture long-range temporal dependencies [64].

Interaction-aware and structured motion modeling

Recognizing that subtle execution differences often arise from structured motion patterns and interactions, some methods go beyond global video features to explicitly model joint relationships, object interactions, or semantic groupings.

Some approaches focuses on joint-level interaction modeling, where local motion patterns around human joints are jointly analyzed with global scene context. By constructing modules that capture both common motion patterns shared across joints and differential motion dynamics among them, these methods learn interaction-aware representations that better reflect coordination and execution quality. Extensions of this idea further introduce adaptive evaluation mechanisms that automatically select or compose different assessment functions for different action categories, enabling action-specific interaction modeling and improving generalization across diverse motion types [68, 69].

Another closely related research direction targets asymmetric interaction modeling in human–object or agent–tool scenarios, which is particularly relevant for tasks involving interactive behaviors such as surgical procedures. In this setting, interactions between primary entities (e.g., humans) and secondary entities (e.g., tools or objects) are modeled asymmetrically to reflect their distinct functional roles. Interaction-aware features are extracted at the segment level and fused with global motion representations for quality regression. More recent work further automates this process by learning to identify primary and secondary entities and performing asymmetric interaction modeling without manual specification, enhancing scalability and robustness [34, 39].

Beyond explicit interaction modeling, several methods adopt multi-path and semantic-structured evaluation frameworks to capture different aspects of action execution. In complex domains such as surgical skill assessment, heterogeneous features-including visual appearance, tool usage, agent behavior, and procedural events-are processed through parallel evaluation paths, whose contributions are adaptively weighted to produce the final assessment. Complementary to this, semantic aggregation strategies dynamically group visual features into semantically meaningful components (e.g., tools, tissue, background) and model temporal dependencies both within and across these groups, enabling fine-grained spatial–temporal reasoning for quality prediction [31, 71].

Structured motion modeling is also extensively explored through graph-based representations, particularly for skeleton-based AQA tasks. Graph convolutional networks operating on spatial–temporal joint graphs are employed to encode structured human motion using multiple information streams, such as joint positions and motion directions. Ensemble and fusion strategies at different levels-data, feature, decision, and model-are designed to effectively integrate complementary motion cues, leading to more robust and expressive skeletal representations. Recent extensions further refine these fusion mechanisms by jointly leveraging positional and directional information in a unified optimization framework [23, 72].

These methods highlight the importance of explicit structural inductive biases in capturing fine-grained execution quality.

Uncertainty-aware and distribution-based score modeling

Most early AQA models formulate score prediction as deterministic regression, implicitly assuming precise ground-truth labels. However, human scoring inherently involves subjectivity and uncertainty, motivating distribution-based formulations.

A representative class of approaches formulates AQA as a score distribution learning problem. Instead of directly regressing a single score, video representations are mapped to a discrete or continuous score distribution, from which the final prediction is derived. Typically, videos are segmented into clips, encoded using shared spatiotemporal feature extractors, and aggregated temporally before applying distribution-based objectives. This formulation enables the model to capture confidence levels across possible scores and has been shown to improve robustness to annotation noise [49, 61]. Extensions of this idea further incorporate temporal alignment mechanisms prior to segmentation, reducing variability caused by temporal misalignment and improving distribution estimation [61].

Another closely related direction introduces latent-variable generative modeling to capture score uncertainty. Variational inference frameworks project video representations into a low-dimensional latent space, where uncertainty is modeled through probabilistic distributions. By sampling from these distributions, models can generate multiple plausible scores that reflect inherent ambiguity in the assessment process. Such uncertainty-aware representations can be fused with global or segment-level visual features to produce final predictions, enabling richer modeling of score variability [63]. More recently, generic distribution auto-encoding modules have been proposed that can be seamlessly integrated into existing AQA architectures, providing a unified mechanism for uncertainty modeling across different backbone designs [66].

Uncertainty-aware and distribution-based methods offer a principled extension to conventional regression-based AQA by embedding probabilistic reasoning directly into the prediction process. By modeling score distributions and latent uncertainty, these approaches better align with the subjective nature of human judgment and complement structural and interaction-aware modeling strategies in modern AQA systems.

Contrastive learning based methods

Contrastive learning was initially introduced as a metric learning objective that models the similarity and dissimilarity between pairs of samples by pulling positive pairs closer and pushing negative pairs apart in the embedding space, providing an effective way to learn discriminative representations [76]. Building upon this idea, later studies developed scalable contrastive learning frameworks for visual representation learning, demonstrating strong performance in image recognition tasks by learning from instance-level comparisons without explicit supervision [77].

In recent years, contrastive learning has emerged as an effective paradigm for AQA, as it naturally aligns with the comparative and ranking-oriented nature of quality evaluation. Unlike conventional AQA approaches, which take a single video as input and directly regress an absolute quality score, pairwise AQA methods operate on a pair of action videos and aim to predict their relative quality or score difference. By emphasizing relative comparison rather than absolute regression, pairwise contrastive AQA methods better reflect the comparative nature of quality judgment and exhibit improved robustness to subjective bias and annotation noise.

Existing contrastive AQA methods can be broadly categorized according to the granularity of comparison, including global video-level contrastive learning (Sect. "Video-level contrastive regression") and stage-aware contrastive comparison strategies (Sec. "Stage-aware and procedure-aware contrastive learning"). This section provides a systematic review of contrastive learning-based AQA methods. The results of these approaches are summarized in Table 5.

Video-level contrastive regression

Early contrastive AQA methods formulate quality assessment as a video-level relative regression problem, where the model predicts the score difference between paired action videos rather than absolute scores.

Table 5 Results of Contrastive Learning Based Methods

Dataset	Method	Publisher	SRCC	R-l_2
AQA-7	CoRe [22]	ICCV 2021	0.8401	2.12
	TPT [17]	ECCV 2022	0.8715	1.68
	PCLN [78]	ECCV 2022	0.8795	-
	T^2CR [79]	Information Sciences 2024	0.8726	1.62
	Uni-FineParser [80]	TPAMI 2025	0.8773	1.72
FineDiving	TSA [37]	CVPR 2022	0.9203	0.342
	MCoRe [81]	ICASSP 2024	0.9232	0.3265
	STSA [82]	IJCV 2024	0.9397	0.2707
	T^2CR [79]	Information Sciences 2024	0.9275	0.3305
	FineParser [80]	CVPR 2024	0.9435	0.2602
	Uni-FineParser [80]	TPAMI 2025	0.9501	0.2294
MTL-AQA	CoRe [22]	ICCV 2021	0.9512	0.260
	TPT [17]	ECCV 2022	0.9607	0.2378
	PCLN [78]	ECCV 2022	0.923	-
	Zhou et al. [43]	TVCG 2023	0.9537	0.245
	T^2CR [79]	Information Sciences 2024	0.9638	0.222
	FineParser [80]	CVPR 2024	0.9585	0.2411
	Uni-FineParser [80]	TPAMI 2025	0.9622	0.2299
JIGSAWS	CoRe [22]	ICCV 2021	0.85	4.556
	TPT [17]	ECCV 2022	0.89	3.668
	T^2CR [79]	Information Sciences 2024	0.91	3.279
RFSJ	TCM [29]	ACM MM 2023	0.9346	0.55
JDM	Zhou et al. [43]	TVCG 2023	0.8147	-

Yu et al.[22] first introduced this paradigm by proposing the Contrastive Regression (CoRe) framework, which processes a target video and a reference video using a shared backbone and predicts their relative score through a group-aware regression module. The final score is obtained by combining the predicted difference with the reference score. By explicitly modeling comparative judgment, CoRe closely mirrors the human evaluation process in sports judging. Subsequent studies explore pairwise contrastive learning strategies integrated into the training process, while retaining standard regression during inference [78]. In this setting, paired video features are first extracted and encoded by a temporal encoder to capture coarse temporal dynamics. The network simultaneously predicts individual scores for each video and learns a relative score constraint through a pairwise contrastive learning objective, which enforces consistency between predicted score differences and ground-truth relative quality. At test time, the model operates on a single video, avoiding the need for reference videos while still benefiting from contrastive supervision during training.

Several extensions further enrich the contrastive formulation by introducing additional supervisory cues or auxiliary inputs. For instance, playback information is incorporated to guide the aggregation of paired video features, where a Temporal Concentration Module selectively emphasizes informative temporal regions during contrastive learning [29]. Other works embed contrastive regression into application-oriented systems, such as combining AQA with motion retrieval and augmented reality visualization. In this case, the predicted contrastive score is used to retrieve and render a corresponding virtual character, enabling intuitive comparison between patient movements and reference examples for clinical assessment [43].

More recent approaches address the limitation that conventional contrastive regression primarily emphasizes difference learning while underutilizing absolute global features. To this end, hybrid frameworks integrate direct regression and contrastive regression within a unified architecture [79]. These methods employ multiple temporal sampling strategies-such as dense and sparse sampling pathways-to extract complementary global representations from the same video. Contrastive learning is applied to align features from different perspectives, enhancing motion consistency modeling, while direct regression predicts absolute scores. The final score is obtained by combining or averaging the outputs of both regression branches, leveraging the strengths of relative comparison and absolute quality estimation.

Despite their effectiveness, video-level contrastive regression methods generally treat actions as indivisible wholes, focusing on global spatio-temporal representations and overall score differences. As a result, they often lack fine-grained temporal modeling and struggle to capture phase-level execution quality or localized errors within complex actions.

Stage-aware and procedure-aware contrastive learning

To overcome the limitations of video-level contrastive regression, a growing body of work introduces stage-aware and procedure-aware contrastive learning, explicitly modeling the internal temporal structure of complex actions. These methods decompose an action into multiple segments, stages, or procedural steps, and perform contrastive learning at finer temporal resolutions to better capture localized execution quality and stage-specific errors.

A representative line of research focuses on stage-aware contrastive modeling, where paired videos are first divided into multiple temporal segments or stages, and contrastive learning is applied within corresponding stages. Typically, segment-level features are extracted using a shared I3D backbone, followed by temporally resolved decoding and aggregation through Transformer-based modules. By employing Transformer decoders with cross-attention mechanisms, these models learn part-level or stage-level temporal representations that emphasize differences between corresponding segments in paired videos. The learning process is often further regularized using ranking-based objectives and sparsity constraints to encourage discriminative and compact representations [17]. An et al. [81] propose multi-stage contrastive regression frameworks explicitly divide actions into predefined or learned stages and compute relative quality scores independently for each stage, which are then aggregated to produce the final prediction.

Beyond stage-level decomposition, procedure-aware contrastive learning treats an action as an ordered sequence of semantically meaningful procedural steps. In this setting, dedicated procedure segmentation modules are used to partition both query and exemplar videos into step-wise representations. The spatiotemporal relationships among these steps are then modeled using procedure-aware cross-attention mechanisms implemented via Transformer decoders, enabling fine-grained alignment and comparison across corresponding steps. The resulting representations are fed into contrastive regression modules to estimate relative quality differences between actions [37, 82]. To further enhance discrimination, Xu et al. [82] introduce spatial-temporal segmentation attention mechanisms to distinguish foreground motion from background context, improving robustness to irrelevant visual content. Xu et al. [83] take the athlete as the core modeling target and explicitly focus on the human body and its key action regions through spatial action parsing, thereby effectively suppressing interference from irrelevant background information.

More recent studies extend procedure-aware contrastive learning toward fine-grained spatial-temporal action parsing, aiming to improve both performance and interpretability. These approaches explicitly parse actions along temporal and spatial dimensions, extracting complementary representations that capture detailed motion patterns and structural cues. Spatial and temporal features are learned separately from paired videos and then integrated to form target action representations, while static visual cues are extracted from the entire video to provide global contextual information. Fine-grained contrastive regression modules, typically built upon Transformer decoders, compute difference scores between the parsed action representations of the query and exemplar videos, enabling more precise quality assessment [80].

Stage-aware and procedure-aware contrastive learning methods significantly enhance the modeling of temporal structure in AQA by aligning and comparing actions at finer granularity. However, these approaches often rely on accurate stage or procedure segmentation and introduce additional architectural complexity, which may limit their scalability and robustness when applied to unconstrained actions or datasets with ambiguous temporal boundaries.

Semi-supervised learning based methods

Fully supervised AQA methods require a large amount of annotated data, which typically demands significant time and effort from professional evaluators, resulting in high costs. In addition, the quality and consistency of the data can be influenced by subjective biases of the evaluators. To reduce this high cost, many researchers have begun exploring the incorporation of semi-supervised learning strategies into AQA tasks, aiming to reduce reliance on manually annotated data. This section provides a review of these semi-supervised learning-based AQA methods, with their results summarized in Table 6.

Zhang et al. [50] combines semi-supervised learning with self-supervised objectives to enhance representation learning. In this paradigm, both labeled and unlabeled videos are processed through a shared encoder to extract intermediate features, while regression is performed only on labeled samples. To exploit unlabeled data, auxiliary self-supervised tasks-such as masked feature reconstruction-are introduced to model temporal dependencies and encourage the encoder to capture meaningful motion patterns. Additionally, adversarial training mechanisms are employed at the feature level to align the distributions of labeled and unlabeled samples, thereby reducing domain discrepancy and improving generalization. A representative class of methods exploits unlabeled data through representation alignment and self-supervised auxiliary objectives. In this paradigm, labeled and unlabeled videos are processed by a shared encoder, while score regression is performed only on labeled samples. Self-supervised tasks, such as masked feature reconstruction or temporal consistency modeling, are introduced to encourage the encoder to capture informative motion patterns from unlabeled data. In addition, feature-level alignment mechanisms, including adversarial training, are employed to reduce distribution discrepancies between labeled and unlabeled samples, thereby improving generalization under limited supervision [50].

More recent approaches extend semi-supervised learning beyond global score regression by incorporating fine-grained temporal and structural modeling. Self-supervised sub-action parsing is introduced to generate pseudo structural annotations for unlabeled videos, enabling the model to learn consistent temporal semantics across labeled and unlabeled data. In parallel, decoupled representation learning frameworks explicitly separate sub-action semantics from execution quality, aligning discrete semantic embeddings and continuous quality representations in a shared embedding space with

Table 6 Results of Semi-supervised Learning Based Methods

Dataset	Method	Publisher	SRCC
MTL-AQA (10% labeled data)	S^4AQA [50]	TCSVT 2022	0.676
	SAP-Net [84]	TIP 2024	0.729
	TRS [85]	ECCV 2024	0.825
MTL-AQA (40% labeled data)	S^4AQA [50]	TCSVT 2022	0.746
	SAP-Net [84]	TIP 2024	0.801
	TRS [85]	ECCV 2024	0.901
	VQD-Net [86]	ICME 2025	0.812
RG (40% labeled data)	S^4AQA [50]	TCSVT 2022	0.342
	SAP-Net [84]	TIP 2024	0.393
	TRS [85]	ECCV 2024	0.529
	VQD-Net [86]	ICME 2025	0.403
JIGSAWS (50% labeled data)	S^4AQA [50]	TCSVT 2022	0.655
	TRS [85]	ECCV 2024	0.753
FineDiving (20% labeled data)	SAP-Net [84]	TIP 2024	0.845
FineFS (50% labeled data)	SAP-Net [84]	TIP 2024	0.709
	VQD-Net [86]	ICME 2025	0.705

confidence-aware pseudo-label supervision. These methods capture detailed temporal structure and improve robustness under limited annotation scenarios [84, 86].

Semi-supervised learning based methods alleviate the dependency on large-scale annotated datasets by effectively leveraging unlabeled videos through representation alignment, consistency regularization, and pseudo-label supervision. By integrating self-supervised objectives, teacher-student paradigms, and fine-grained structural cues, these approaches enhance both feature learning and score prediction under limited supervision, offering a practical and scalable solution for real-world AQA scenarios where expert annotations are scarce.

Domain shift-aware methods

Domain shift-aware AQA methods explicitly address the mismatch between action recognition pre-training and the fine-grained requirements of quality assessment. By refining representations through coarse-to-fine modeling, intermediate domain adaptation, or progressive feature-level alignment, these approaches reduce reliance on limited annotations and improve robustness and generalization. Such strategies are particularly effective in small-scale or long-term AQA scenarios, highlighting domain adaptation as a critical direction for advancing practical and reliable assessment. The results of the methods reviewed in this section are summarized in Table 7.

A representative class of approaches reduces domain mismatch through coarse-to-fine representation refinement inspired by human judging processes. These methods decompose quality assessment into multiple granularity levels, such as coarse grading followed by fine-grained evaluation, and explicitly align representations across these levels. By parsing video features into hierarchical or instruction-aware components and integrating them for final score prediction, such approaches alleviate task mismatch and improve robustness under limited supervision [87, 88]. Progressive refinement strategies further enable feature adaptation in a shallow-to-deep manner without requiring full backbone fine-tuning, making them particularly suitable for long-term AQA scenarios with constrained annotation budgets.

Another line of work focuses on intermediate domain adaptation to bridge the gap between action recognition pre-training and AQA objectives. Instead of directly fine-tuning on scarce labeled AQA data, these methods introduce an additional adaptation stage that leverages unlabeled in-domain videos. Through feature-level alignment during this intermediate stage, spatiotemporal representations are gradually shifted toward quality-sensitive characteristics, effectively narrowing the domain gap and improving generalization across datasets and scenarios [20].

Table 7 Results of Domain Shift-aware Methods

Dataset	Method	Publisher	SRCC	R-l_2
MTL-AQA	PECoP [20]	WACV 2024	0.952	-
FineDiving	PECoP [20]	WACV 2024	0.9315	-
JIGSAWS	PECoP [20]	WACV 2024	0.89	-
PD4T	PECoP [20]	WACV 2024	0.6387	-
LOGO	PHI [87]	TIP 2025	0.835	2.752
RG	CoFInAl [8]	IJCAI 2024	0.807	-
	PHI [87]	TIP 2025	0.810	2.300
Fis-V	CoFInAl [8]	IJCAI 2024	0.788	-
	PHI [87]	TIP 2025	0.804	2.178

Overall, domain shift-aware AQA methods emphasize adapting pre-trained representations to the specific demands of quality assessment rather than relying solely on task-level supervision. By incorporating coarse-to-fine refinement and intermediate domain alignment, these approaches reduce dependence on large-scale annotations and enhance robustness, particularly in small-scale or long-duration AQA settings. This highlights domain adaptation as a critical direction for advancing practical and reliable assessment.

Continual learning based methods

Most existing AQA methods assume a static training setting, where all action types and data distributions are available upfront. However, real-world AQA systems often face non-stationary data streams and evolving assessment tasks, making continual learning [89] a critical yet underexplored problem. Recent studies have begun to introduce continual learning paradigms into AQA, aiming to mitigate catastrophic forgetting while preserving score-discriminative representations across sequential tasks. The results of the methods reviewed in this section are summarized in Table 8. Notably, in Table 8, ρ_{avg} denotes the overall correlation [89], which differs from the conventional SRCC. Specifically, ρ_{avg} is designed for continual learning scenarios by aggregating samples from all observed tasks and computing a unified rank correlation, thereby providing a more robust and fair evaluation under non-stationary and task-incremental settings.

Zhou et al. [90] are the first to formally introduce continual learning into AQA, addressing a fundamental limitation of existing AQA methods that assume access to all training data at once and suffer from severe catastrophic forgetting when action assessment tasks are learned sequentially. They formulate a Continual AQA (CAQA) setting that enables models to incrementally adapt to evolving skills under non-stationary data streams, while operating under strict privacy constraints that prohibit storing raw historical samples. A key challenge identified in this setting is the misalignment between replayed historical features and the continuously evolving feature manifold induced by backbone updates. To tackle this issue, the authors propose Manifold-Aligned Graph Regularization (MAGR), which aligns historical features to the current feature manifold and enforces graph-based consistency between feature distributions and continuous quality score distributions. By explicitly addressing feature manifold drift in a regression-based continual learning scenario, this work establishes CAQA as a practically motivated and technically distinct extension of continual learning to the AQA domain.

Table 8 Results of Continual Learning Based Methods

Dataset	Method	Publisher	SRCC	ρ_{avg}
MTL-AQA	MAGR [90]	ECCV 2024	-	0.8979
	PECoP [20]	WACV 2024	0.952	-
	MAGR++ [91]	ARXIV 2025	-	0.9383
UNLV-Diving	MAGR [90]	ECCV 2024	-	0.7668
	MAGR++ [91]	ARXIV 2025	-	0.8165
FineDiving	MAGR [90]	ECCV 2024	-	0.858
	PECoP [20]	WACV 2024	0.9315	-
	MAGR++ [91]	ARXIV 2025	-	0.8902
JDM	MAGR [90]	ECCV 2024	-	0.7166
JIGSAWS	PECoP [20]	WACV 2024	0.89	-
PD4T	PECoP [20]	WACV 2024	0.6387	-

Building upon this formulation, subsequent studies explore complementary directions to improve scalability and long-term stability in continual AQA. One class of methods emphasizes parameter-efficient adaptation, introducing lightweight adapter or prompt-like modules into pretrained video backbones. By updating only a small subset of parameters during continual learning, these approaches significantly reduce computational and storage overhead while mitigating catastrophic forgetting, making continual adaptation feasible for long-term and resource-constrained AQA systems [20]. Another line of work focuses on maintaining score-discriminative consistency across sequential tasks. Rather than treating tasks as independent, these methods emphasize preserving the correlation between learned representations and quality scores over time. By explicitly modeling general and task-specific assessment knowledge and enforcing feature–score consistency, such approaches achieve more stable performance under sequential learning and further consolidate continual AQA as a unified and principled learning framework [91].

Continual learning based AQA methods extend quality assessment from static, closed-world settings to realistic, evolving scenarios. By addressing feature manifold drift, parameter efficiency, and score consistency, these approaches lay the groundwork for practical lifelong AQA systems capable of adapting to new tasks while retaining reliable assessment performance.

Causality-based and rule-aware methods

As AQA tasks are increasingly applied in professional sports analytics and intelligent judging systems, researchers have gradually recognized that black-box models relying solely on end-to-end regression are insufficient to meet the requirements for robustness, fairness, and interpretability. In recent years, a line of work has begun to incorporate causal modeling, causal attention mechanisms, and explicit rule-based constraints to mitigate background interference, temporal bias, and inconsistencies in subjective scoring. This section summarizes related studies from two perspectives: causal-aware approaches and rule-based interpretable methods. The quantitative results of these methods are summarized in Table 9.

Causality-aware AQA methods aim to go beyond purely correlational learning by explicitly modeling causal relationships among video features, action stages, and quality scores. A representative line of work formulates AQA from a causal inference perspective by constructing explicit causal graphs that connect video-level representations, stage-wise semantics, and final assessment scores, and by introducing intervention mechanisms to suppress spurious background correlations while emphasizing human-centric causal cues [53]. Within this framework, graph-based causal modeling and temporal causal attention are jointly employed to disentangle foreground action semantics

Table 9 Results of Causality-based and Rule-aware Methods

Dataset	Method	Publisher	SRCC	R-l_2
MIT-Skating	IRIS [92]	IUI 2023	0.829	-
UNLV-Diving	TFE+FCCA [93]	SENSORS 2025	0.882	-
FineDiving	FineCausal [53]	CVPRW 2025	0.9447	0.2338
RG	CaFlow [94]	ARXIV 2025	0.841	2.449
Fis-V	CaFlow [94]	ARXIV 2025	0.805	2.05
LOGO	CaFlow [94]	ARXIV 2025	0.856	1.429

from confounding context and to enable fine-grained attribution of sub-actions to the final score, thereby improving both score discrimination and interpretability.

Building upon this causal formulation, subsequent studies extend causality-aware modeling to more challenging long-duration and high-dynamic AQA scenarios, where temporal instability and motion ambiguity are prominent. These methods incorporate cross-modal representations and long-term temporal reasoning to enhance causal robustness in complex sports actions [93, 94]. Specifically, dual-stream architectures leveraging RGB and optical flow are adopted to capture complementary motion cues, while temporal feature enhancement and causal cross-modal attention mechanisms expand temporal receptive fields and preserve causal ordering in long sequences. In addition, counterfactual regularization and bidirectional time-conditioned modeling are introduced to explicitly disentangle causal features from confounding factors and to stabilize long-term temporal representations, resulting in more robust and consistent performance in long-duration AQA tasks.

In parallel to statistical causality-based approaches, another line of research focuses on improving interpretability by explicitly modeling scoring rules and symbolic reasoning processes. These methods embed domain knowledge and official judging criteria directly into the AQA pipeline, enabling transparent and traceable score generation [92, 95]. Typically, neural networks are first employed to extract low-level visual features or fundamental symbols from videos, which are then transformed into higher-level semantic or geometric representations. Based on predefined scoring rubrics, actions are segmented into distinct phases and evaluated through rule-based aggregation or symbolic reasoning, producing structured, phase-level assessment results. By aligning model outputs with human-understandable scoring logic, such rule-based and neuro-symbolic AQA methods provide more objective, explainable, and trustworthy evaluation outcomes.

Causality-aware and rule-driven approaches address complementary limitations of conventional black-box AQA models. While causal modeling improves robustness and generalization by suppressing non-causal correlations and stabilizing temporal reasoning, rule-based and neuro-symbolic methods enhance transparency by making the evaluation process explicitly interpretable. Together, these directions advance AQA toward more reliable, explainable, and practically deployable assessment systems.

Multi-modal based methods

Most researchers implementing AQA tasks rely primarily on single-modality information, such as RGB video. Although this approach is straightforward, it often provides limited information. RGB videos primarily capture color and luminance details; however, for assessing complex action quality, these videos may not adequately capture motion, depth, or fine action details. Consequently, an increasing number of researchers are exploring multi-modal information fusion to enhance the performance of the AQA system. Multi-modal fusion uses data from various sensors and sources, such as skeleton data, optical flow, and audio information. These additional modalities provide a richer context, enabling AQA systems to achieve a more comprehensive understanding of each aspect of an action.

To provide a structured review, we categorize multi-modal AQA methods according to the functional role played by different modalities in quality assessment. Specifically, we first review approaches that combine visual and pose information to enhance motion representation and structural modeling (Sect. "Motion and structural enhancement via visual and pose modalities"), thereby improving the characterization of fine-grained execution quality. We then discuss methods that introduce semantic-level multi-modal supervision (Sect. "Semantic-level multi-modal supervision and interpretability"), such as textual descriptions or rule-based annotations, to improve interpretability and provide higher-level guidance for quality evaluation. Finally, we review audio-visual fusion methods (Sect. "Audio-visual multi-modal fusion for AQA") that exploit acoustic cues, such as rhythm and timing, as complementary signals to visual information for assessing action quality. The results of the discussed approaches summarized in Table 10.

Motion and structural enhancement via visual and pose modalities

To enhance motion understanding and structural modeling in AQA, a line of research explores the integration of pose-based representations with visual cues from RGB videos or optical flow. These data provide explicit structural and kinematic information that is often complementary to appearance-based features.

Many sports-oriented AQA datasets only provide RGB videos, and pose estimation [104, 105] under extreme motion conditions-such as diving or gymnastics-remains challenging due to severe self-occlusion and highly contorted body configurations. To

Table 10 Results of Multi-modal Based Methods

Dataset	Method	Modality	Publisher	SRCC	R-l_2	Accuracy
MTL-AQA	SGN [26]	RGB+Text	TMM 2023	0.9607	0.232	-
	VATP-Net [96]	RGB+Text	TCSVT 2024	0.9588	-	-
	Xu et al. [97]	RGB+Flow	MMSP 2024	0.9412	0.357	-
	NAE [51]	RGB+Text	CVPR 2024	0.943	0.34	-
	RICA^2 [98]	RGB+Text	ECCV 2024	0.962	0.228	-
	MVLA [99]	RGB+Text	ECCV 2024	0.9655	0.224	
AQA-7	EAGLE-Eye [100]	RGB+Skeleton	WACV 2021	0.814	-	-
	2M-AF [101]	RGB+Skeleton	ACM MM 2024	0.8662	-	-
UNLV-Diving	FALCONS [102]	RGB+Skeleton	CVPR 2020	0.8453	-	-
	2M-AF [101]	RGB+Skeleton	ACM MM 2024	0.8838	-	-
MIT-Skating	EAGLE-Eye [100]	RGB+Skeleton	WACV 2021	0.601	-	-
FS1000	SGN [26]	RGB+Text	TMM 2023	0.847	-	-
	Skating-Mixer [42]	RGB+Audio	AAAI 2023	0.8186	-	-
	MLAVL [103]	RGB+Text+Audio	CVPR 2025	0.90	-	-
Fis-V	SGN [26]	RGB+Text	TMM 2023	0.765	-	-
	Skating-Mixer [42]	RGB+Audio	AAAI 2023	0.75	-	-
	PAMFN [27]	RGB+Flow+Audio	TIP 2024	0.822	-	-
	MVLA [99]	RGB+Text	ECCV 2024	0.783	-	-
	MLAVL [103]	RGB+Text+Audio	CVPR 2025	0.823	-	-
OlympicFS	SGN [26]	RGB+Text	TMM 2023	0.9159	-	-
RG	PAMFN [27]	RGB+Flow+Audio	TIP 2024	0.819	-	-
	VATP-Net [96]	RGB+Text	TCSVT 2024	0.8	-	-
	MLAVL [103]	RGB+Text+Audio	CVPR 2025	0.849	-	-
AGF-OLYMPICS	DNLA [25]	RGB+Skeleton	TIM 2024	0.68	-	-
FineGym	NAE [51]	RGB+Text	CVPR 2024	0.749	1.55	-
PISA	Parmar et al. [13]	RGB+Audio	MMSP 2021	-	-	0.746
FineFS	VATP-Net [96]	RGB+Text	TCSVT 2024	0.7725	-	-
FineDiving	MVLA [99]	RGB+Text	ECCV 2024	0.9419	0.284	-
JIGSAWS	RICA^2 [98]	RGB+Text	ECCV 2024	0.92	-	-
	MVLA [99]	RGB+Text	ECCV 2024	0.9	2.964	-

address pose extraction under extreme sports scenarios, Nekoui et al. [102] introduce specialized datasets and pose-aware AQA frameworks. By constructing extreme-action pose datasets and training dedicated pose estimators, more reliable skeletal representations can be obtained for downstream assessment. These approaches typically model spatiotemporal dependencies in skeleton sequences using graph-based networks, while appearance cues are extracted from RGB videos using lightweight 3D CNN architectures. The two modalities are fused via task-aware weighting or bridging mechanisms to estimate execution quality scores. This line of work was further extended to cover a broader range of extreme sports by expanding pose datasets and introducing modality-specific temporal modeling modules, where fine-grained temporal dynamics are emphasized in skeleton streams and coarser temporal patterns are captured from RGB videos before multimodal fusion [100].

Beyond dataset-driven pose estimation, some methods focus on designing general multimodal fusion frameworks that jointly leverage skeletal and visual information [25, 101]. In such approaches, skeletal features are commonly extracted using graph convolutional networks, sometimes enhanced with self-supervised masking strategies to improve robustness, while RGB videos are segmented and processed using 3D CNNs to obtain segment-level or global representations. Fusion modules then integrate modality-specific predictions or features, either by learning modality preferences or adaptively selecting the more reliable stream for final score estimation. This design allows the model to dynamically balance the contributions of pose and appearance cues depending on their respective reliability.

In addition to skeleton–RGB fusion, motion enhancement has also been explored through the integration of RGB video and optical flow modalities. In this setting, appearance and motion features are extracted independently and subsequently enhanced using Transformer-based encoders. Cross-attention mechanisms are then applied to explicitly model interactions between motion and appearance features, enabling the network to better align dynamic motion cues with visual context before score regression [106]. Related works further extend this paradigm by adopting paired-sample learning and multi-stage fusion pipelines, where multimodal feature enhancement, temporal segmentation, contrastive comparison, and adaptive aggregation are jointly employed to capture fine-grained motion differences and improve scoring accuracy [97].

Beyond skeleton-RGB fusion, some studies enhance feature representations by integrating RGB videos with the optical flow modality. In this setting, appearance and motion features are extracted independently and subsequently enhanced using Transformer-based encoders. Cross-attention mechanisms are then applied to explicitly model interactions between motion and appearance features, enabling the network to better align dynamic motion cues with visual context before score regression [106]. Xu et al. [97] further extend this paradigm by adopting paired-sample learning and multi-stage fusion pipelines, where multimodal feature enhancement, temporal segmentation, contrastive comparison, and adaptive aggregation are jointly employed to capture fine-grained motion differences and improve scoring accuracy.

Motion and structural enhancement methods leverage complementary visual and pose-based modalities to provide a more explicit and interpretable representation of action quality. By integrating appearance cues from RGB or optical flow with structured motion information from skeletal data, these approaches improve the modeling of fine-grained kinematics, long-range temporal dependencies, and motion consistency. Multimodal fusion strategies-ranging from graph-based skeleton modeling to attention-driven cross-modal interaction-enable AQA systems to better capture both how an action is performed and how well its motion aligns with expected execution patterns, making them particularly effective for complex, dynamic, and long-duration actions.

Semantic-level multi-modal supervision and interpretability

An emerging line of research introduces semantic-level multi-modal supervision into AQA, aiming not only to improve scoring accuracy but also to enhance interpretability and user-oriented feedback. These methods typically leverage textual information-such as expert commentary, semantic annotations, or automatically generated descriptions-as auxiliary supervision to guide visual representation learning and to ground score prediction in explicit semantic concepts.

A representative group of approaches incorporates textual descriptions or expert commentary as auxiliary semantic signals during training, while retaining numerical score regression as the primary objective [26, 96, 99]. In this paradigm, textual information is aligned with visual features at different temporal or semantic granularities, such as sub-action segments, action phases, or holistic performance descriptions. By encouraging consistency between visual representations and corresponding semantic cues, these methods enable semantically grounded feature learning and improve the interpretability of temporal modeling. Notably, language supervision is often used only during training, allowing inference to rely solely on visual inputs without additional annotation requirements.

Beyond auxiliary supervision, some works explicitly exploit language to model sub-action structures and scoring composition [98]. In these approaches, textual descriptions-often generated by large language models-are used to decompose complex actions into stepwise sub-actions, which are then fused with clip-level visual features through cross-attention mechanisms. Sub-action–level score distributions are predicted and subsequently aggregated using structured formulations, such as graph-based modeling, enabling rule-aware score composition and more transparent attribution of performance quality.

Moving further toward user-facing interpretability, another line of research reframes AQA as a joint scoring-and-explanation problem by directly generating textual feedback alongside numerical scores [51]. These methods emphasize narrative-level assessment, where prompt-guided multi-modal interaction frameworks are designed to couple linguistic prompts with visual representations. As a result, models are capable of producing descriptive explanations that articulate performance strengths and weaknesses in natural language, offering richer and more accessible feedback than scalar scores alone.

Semantic-level multi-modal AQA methods leverage textual information as high-level semantic supervision or explicit explanatory output, enabling models to associate visual patterns with meaningful action concepts. Compared with purely regression-based approaches, this paradigm improves robustness and interpretability by grounding score prediction in language-aligned semantics and structured descriptions, highlighting a promising direction toward explainable and user-oriented action assessment systems.

Audio-visual multi-modal fusion for AQA

Beyond visual cues alone, a line of AQA research incorporates audio information to complement visual representations, motivated by the observation that sound and music convey valuable cues related to rhythm, timing, and performance quality. This direction is particularly relevant for actions where execution quality is tightly coupled with auditory patterns, such as musical performance and artistic sports.

Early audio-visual AQA studies focus on jointly modeling visual motion cues and audio signals to capture complementary aspects of performance quality [13, 42]. In these approaches, visual features encode body movements and spatial–temporal dynamics, while audio features provide information about rhythm, tempo, and synchronization. By fusing the two modalities at the feature or temporal level, the models obtain a more holistic representation of performance quality than visual-only methods, demonstrating improved robustness in scenarios where timing and rhythmic consistency are critical for scoring.

Subsequent works extend audio-visual fusion by incorporating higher-level semantic or structural guidance [103]. In this setting, domain-specific knowledge-often derived from textual resources-is used to construct action knowledge graphs or semantic priors, which guide audio and visual features to attend to key action elements and their alignment with musical rhythm or scoring rules. Such designs strengthen modality interaction by explicitly encoding domain constraints, enabling more targeted and interpretable audio-visual reasoning.

More recent approaches further explore adaptive and multi-source fusion strategies to better exploit complementary modalities [27]. Instead of relying on fixed fusion schemes, these methods progressively integrate RGB video, optical flow, and audio signals, allowing the model to dynamically refine cross-modal interactions and suppress redundant or less informative cues. This adaptive fusion paradigm enhances flexibility and improves performance across diverse action types and recording conditions.

Audio-visual multi-modal AQA methods leverage auditory cues to enrich visual representations, leading to more comprehensive and robust quality assessment. They are particularly effective in scenarios where action execution is closely synchronized with sound or music, as they better capture rhythmic structure, temporal consistency, and cross-modal coherence that are difficult to infer from visual information alone.

Summary and discussion

In this section, we reviewed existing AQA methods from multiple methodological perspectives, ranging from traditional handcrafted approaches to modern learning paradigms such as contrastive, semi-supervised, continual, and multi-modal learning. These methods differ significantly in their modeling assumptions, data requirements, robustness, and practical applicability. To facilitate a clearer understanding, we summarize the characteristics of each category and discuss the problem settings in which they are most suitable.

Traditional AQA methods (Sect. "Traditional methods") typically rely on handcrafted features and explicit temporal alignment or rule-based scoring schemes. While their performance is generally limited compared to learning-based approaches, they offer strong interpretability and require minimal training data, making them suitable for constrained scenarios with limited annotations or strong domain priors. General deep learning–based methods (Sect. "General deep learning methods") model AQA as a supervised regression or classification problem and achieve strong performance under sufficient labeled data. However, these methods often assume closed-world settings and may struggle with data scarcity, domain shift, and long-term deployment. To alleviate the reliance on large-scale labeled datasets, contrastive learning-based methods (Sect. "Contrastive learning based methods") and semi-supervised approaches (Sect. "Semi-supervised learning based methods") have been proposed. Contrastive methods emphasize representation learning by exploiting relational or paired supervision, which is particularly effective when relative quality information is available. Semi-supervised methods, on the other hand, leverage unlabeled or weakly labeled data and are well suited for real-world scenarios where annotation costs are prohibitive. Domain shift-aware methods (Sect. "Domain shift-aware methods") explicitly address performance degradation caused by distribution mismatch across datasets, environments, or subjects, making them suitable for cross-dataset evaluation and deployment in open-world settings. Continual learning-based methods (Sect. "Continual learning based methods") further extend this capability by enabling models to incrementally adapt to new data while mitigating catastrophic forgetting, which is essential for long-term training or monitoring systems. Causality-based and rule-aware methods (Sect. "Causality-based and rule-aware methods") aim to incorporate explicit structural knowledge, causal reasoning, or domain rules into the assessment process. Although such methods may sacrifice some flexibility, they provide improved interpretability and controllability, which are critical in high-stakes applications such as sports judging, rehabilitation, and skill training. Multi-modal AQA methods (Sect. "Multi-modal based methods") integrate complementary information from pose, audio, textual, or semantic modalities, enhancing robustness and expressiveness. These approaches are particularly effective in complex actions where visual appearance alone is insufficient to capture quality variations.

Overall, the choice of AQA methodology should be guided by the specific application requirements, including data availability, domain variability, interpretability demands, and deployment duration. While supervised deep learning methods remain dominant under controlled settings, emerging paradigms such as contrastive, continual, causal, and multi-modal learning provide promising solutions for more realistic and challenging AQA scenarios.

Application of AQA in sports

In the field of sports, AQA technology is primarily employed to evaluate and enhance performance in technically demanding disciplines such as diving, rhythmic gymnastics, and figure skating. These actions require strict standards for both precision and artistry in performance, alongside fair and detailed scoring criteria. Traditional manual scoring methods are often influenced by subjective factors; however, AQA technology, utilizing advanced computer vision and deep learning models, offers more objective and accurate scoring and feedback. The application of AQA technology in sports can be broadly summarized in two aspects: first, it is used in competitions to provide quantitative scoring of movements, ensuring the objectivity and fairness of the results; second, it generates targeted feedback to help athletes identify areas for improvement, enabling focused training to enhance their skill levels. To further clarify the development of AQA applications in sports, we structure our discussion around two key perspectives. Figure 8 shows the overall structure of this section.

Competition scoring

In competitive sports, scoring mechanisms vary significantly across disciplines. In sports such as running and swimming, performance evaluation is fully objective and determined by measurable outcomes, such as completion time. In contrast, sports including diving, gymnastics, figure skating, and synchronized swimming rely on referee-based scoring systems, where athletes are evaluated according to technical execution, artistic expression, and rule-specific criteria. Although standardized judging rules are established, the scoring process inevitably involves subjective factors, leading to potential inconsistencies and bias. Within this context, AQA has emerged as an enabling technology to support competition scoring by providing objective, fine-grained, and reproducible evaluations of athletic performance from video data.

In this survey, we view competition scoring as a unified application scenario in which AQA methods are developed to support referee-oriented evaluation. In practice, competition scoring encompasses a broad spectrum of assessment settings, including short- and long-duration routines, single-athlete and multi-person interactions, as well as coarse- and fine-grained scoring criteria. Accordingly, we systematically review existing AQA applications in competition scoring-primarily centered on Olympic sports-by organizing them along three key dimensions: action duration, participant structure, and evaluation granularity. This hierarchical taxonomy makes the relationships among these scenarios explicit and aligns the survey with real-world competition workflows, thereby improving clarity and navigability for readers.

Short-term action assessment

Pirsiavash et al. [9] collected an MIT dataset that includes diving and figure skating events. They trained a regression model using handcrafted features to predict action

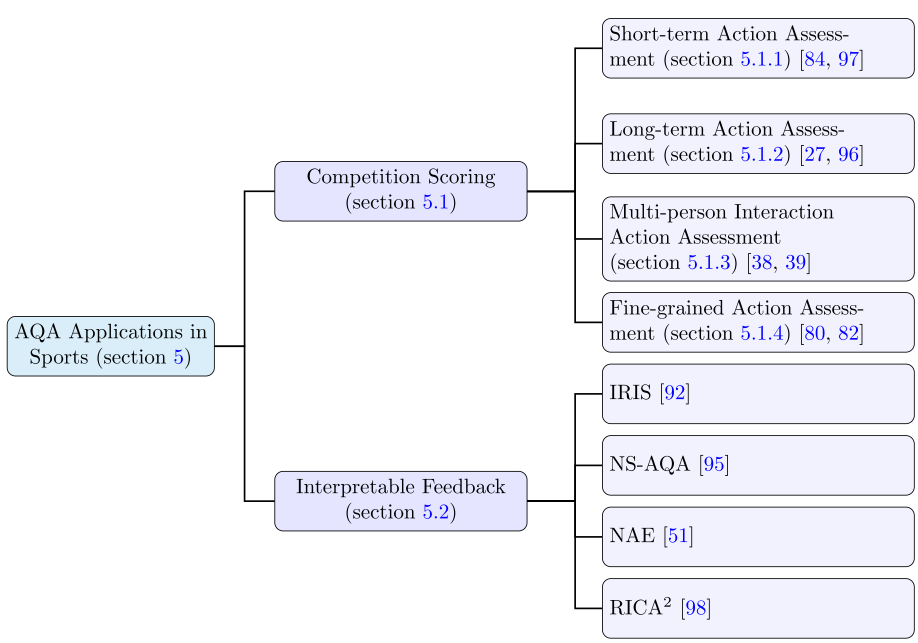

Fig. 8 Overview of AQA applications in sports

scores. In the case of diving, their method achieved a SRCC of 0.41. From the results, it is evident that the handcrafted feature approach showed a significant gap when compared to the scores given by professional judges. Parmar et al. [7] extended the MIT-Dive dataset and additionally collected a dataset for gymnastics vault, named UNLV-Vault. Both gymnastics vault and diving are short-term sports. They segmented the video into 16-frame clips and used a 3D ConvNets to extract features from each clip. After feature fusion, they predicted scores through a regression model. This deep learning approach showed significant improvements compared to the handcrafted feature method. On the UNLV-Dive dataset, the SRCC reached 0.78, and on the smaller-scale MIT-Dive dataset, it achieved 0.74. For the UNLV-Vault dataset, the SRCC reached 0.66. [16] refined the scoring prediction network based on this method by incorporating multiple convolution and pooling operations, followed by a fully connected layer to compute the scores. The model was optimized using a combination of MSE and ranking loss. As a result, their proposed method achieved an improvement of approximately 2 percentage in SRCC on both the UNLV-Dive and UNLV-Vault datasets compared to the method proposed by parmar et al. [7]. On the MIT-Skating dataset, the performance improved by approximately 4 percentage. Xiang et al. [48] introduced ED-TCN [107] for segmenting diving action videos. This segmentation approach better captures the different phases of the entire diving movement. Subsequently, multiple P3D modules were used to extract features from the segments, and score predictions were made after feature fusion. Ultimately, their method achieved a SRCC of 0.86 on the UNLV-Dive dataset.

Parmar et al. [33] collected a diving dataset called MTL-AQA. The dataset includes scores, difficulty levels, and movement description labels for each sample. They improved AQA performance using a multitask learning approach, with their proposed C3D-AVG-MTL method showing improved performance on both the MTL-AQA and UNLV-Dive datasets compared to single-task learning methods. Parmar et al. [32] collected the AQA-7 dataset, which includes multiple types of sports. They conducted experiments on this dataset using the AQA model proposed by parmar et al. [7]. Pan et al. [68] performed the AQA task using full videos and localized video segments around the athlete's joints, validating the approach in the AQA-7 dataset. This method performed worse in diving compared to the approach proposed by parmar et al. [7], but outperformed previous methods in other sports, achieving an average SRCC of 0.7849. Building upon this method, pan et al. [69] introduced an adaptive action evaluation approach, which designs different evaluation architectures tailored to various types of movement. This approach led to a significant performance improvement in the AQA-7 dataset, achieving an average SRCC of 0.85.

Wang et al. [57] extracted spatiotemporal features from videos using SCN and TCN, and fused these features through an attention mechanism. The proposed method achieved significant improvements on the MIT-Skating and UNLV-Vault datasets, with SRCC reaching 0.71 and 0.76, respectively. On the UNLV-Dive dataset, it achieved a SRCC of 0.85, which is approximately one percentage lower than the method proposed by xiang et al. [48]. The results suggest that the proposed framework has a significant advantage in handling long-term actions. Wang et al. [35] focused on feature extraction from videos and proposed TSA-Net, which enables the model to extract features more aligned with the needs of the AQA task. Their method achieved a SRCC of 0.8476 on AQA-7 and 0.9393 on MTL-AQA. Nagai et al. [62] improved the training strategy and introduced a new loss function to enhance model performance. The proposed method showed improvements over the approach proposed by parmar et al. [33] on the MTL-AQA dataset.

Some researchers have incorporated unsupervised and semi-supervised learning into AQA tasks, using a small amount of labeled data to train the models, thereby reducing the annotation overhead. Zhang et al. [50] combined self-supervised learning and unsupervised learning for the AQA task. Their method was validated on the MTL-AQA dataset, where they trained using 10% and 40% labeled data. When using 40% labeled data, they achieved a SRCC of 0.746, which represented an improvement of about 10 percentage compared to training with 10% labeled data. Gedamu et al. [84] proposed SAP-Net, which constructs a teacher-student network to learn consistent action features and subaction representations through self-supervised learning and contrastive learning. Their method was validated on multiple sports datasets, demonstrating improved results compared to several existing AQA methods. Yun et al. [85] proposed a semi-supervised learning method for AQA tasks, using a teacher-student architecture. Their method achieved a SRCC of 0.901 when trained on 40% labeled data from the MTL-AQA. This semi-supervised approach outperformed some fully supervised methods. Given the labor-intensive and costly nature of collecting annotated data for AQA tasks, this semi-supervised and unsupervised approach is particularly well-suited for such tasks.

Yu et al. [22] introduced contrastive learning into AQA, proposing the CoRe framework, which performs the AQA task by regressing the relative scores between the input video and reference video. This approach provides a novel perspective on the AQA problem. In addition, they introduced a more stringent evaluation metric for AQA tasks, called relative L2-distance (R-ℓ_2). Their proposed method achieved an average SRCC of 0.8401 and an average R-ℓ_2 of 2.12 on the AQA-7 dataset. This innovative approach has had a profound impact on subsequent research, with many researchers adopting contrastive learning methods to enhance the accuracy of AQA models [17, 37, 78, 80–82].

In sports scoring, the final score is typically given by multiple judges. After removing the highest and lowest scores, the sum of the remaining scores becomes the final score. This scoring method introduces uncertainty into the final score. Additionally, subjective factors in the judges' evaluations can further contribute to this uncertainty. Many researchers have designed models to address and simulate this uncertainty in scoring. The USDL method proposed by tang et al. [49] accomplishes the AQA task by predicting the probability distribution of scores. This method achieved improvements across various sports in the AQA-7 dataset, with an average SRCC of 0.8102 for all sports. In diving, the scores are given by multiple judges and the final score is calculated based on specific rules. To simulate this scoring method, they introduced the MUSDL approach, which uses multiple USDL models to simulate the judgments of multiple referees. This method achieved a SRCC of 0.9273 on the MTL-AQA dataset. The UD-AQA method, proposed by zhou et al. [63], generates multiple scores to simulate the evaluations of multiple judges for an action. This approach achieved a SRCC about 0.0033 higher than CoRe on the MTL-AQA dataset. While CoRe provides a single final predicted score, UD-AQA simulates the uncertainty in scoring by offering multiple scores, which better aligns with real-world scoring practices. Zhang et al. [66] introduced a regression module, DAE, to capture uncertainty in video features. They embedded this module into the CoRe framework, demonstrating improvements on both the AQA-7 and MTL-AQA datasets.

Some researchers have improved the performance of AQA models by integrating multi-modal information. Nekoui et al. [102] utilized a pose estimation model to extract the skeleton information of athletes, which was then fused with RGB video to perform the AQA task. Their method achieved a SRCC of 0.8453 on the UNLV-Dive dataset. Although their approach showed slightly lower performance compared to the method by Xiang et al. [48], it provided valuable insights for multi-modal AQA methods. [97] proposed a method for evaluating diving movements using RGB videos and optical flow. They validated the effectiveness of incorporating optical flow information in assessment through experiments on the FineDiving and MTL-AQA datasets.

Long-term action assessment

Long-term actions, such as figure skating and rhythmic gymnastics, involve longer action cycles, where athletes must demonstrate high levels of technical and artistic performance over more complex environments and extended periods of time. Evaluating long-duration actions requires not only precise capture of the athlete's technical movements but also a comprehensive consideration of factors such as continuity and rhythm. The evaluation of these actions is more challenging compared to short-duration actions.

The handcrafted feature-based approach proposed by Pirsiavash et al [9]. achieved a SRCC of only 0.35 on the MIT-Skating dataset. This performance is lower than that on the MIT-Dive dataset, indicating that handcrafted features are less effective for long-term actions. Parmar et al. [7] used 3D ConvNets to extract features, and their method achieved a SRCC of 0.53 on the MIT-Skating dataset, marking a significant improvement over the handcrafted feature-based approach. However, this result still lags behind the performance on short-term actions. The MIT-Skating dataset only contains total score labels, which are the sum of TES and PCS. Xu et al. [8] collected the FisV dataset, where each sample is labeled with both TES and PCS scores. They developed S-LSTM and M-LSTM modules to further analyze the features extracted from figure skating videos. The features extracted by both modules are aggregated to predict the final score. This method achieved a SRCC of 0.59 on the MIT-Skating dataset, showing an improvement over the methods proposed by parmar et al. [7] and Li et al. [16].

Zeng et al. [28] collected a dataset called Rhythmic Gymnastics, which includes gymnastics events involving the ball, clubs, hoop and ribbon. They used information from dynamic video segments and static image frames to predict movement scores. Their proposed method was tested on the MIT-Skating and Rhythmic Gymnastics datasets, both of which involve long-term actions. The method achieved a Spearman correlation of 0.615 on the MIT-Skating dataset and an average SRCC of approximately 0.618 for the four types of events in the Rhythmic Gymnastics dataset. Xu et al. [60] proposed the GDLT framework based on the Transformer architecture. This method achieved a SRCC of 0.765 in the Rhythmic Gymnastics dataset and an average SRCC of 0.761 in the Fis-V dataset. The result also indicated that the Transformer architecture has certain advantages in handling long-term actions. Liu et al. [29] used contrastive learning to regress relative scores, while utilizing replay video information to guide model training. To facilitate this, they collected a new figure skating dataset, RFSJ, which includes replay videos. They conducted experiments with several existing AQA methods on this dataset. Their method outperformed other approaches on the RFSJ dataset.

AQA models typically perform worse on long-term actions compared to short-term actions. This may be due to the fact that long action sequences exceed the range of effective information that RGB videos can provide, limiting the model's ability to capture long-term information. To address this issue, some researchers have explored the use of multi-modal information to enhance the model's ability to understand long-term actions, thereby improving performance in such cases. Nekoui et al. [100] used a similar approach to that of their earlier work [102] by integrating skeletal and video information. They validated their method on the AQA-7 and MIT-Skating datasets, achieving Spearman correlations of 0.8140 and 0.6010, respectively. In addition, several other multi-modal methods have been proposed, which integrate audio, text, and other modalities with RGB video to perform AQA tasks. Xia et al. [42] collected a figure skating dataset, FS1000, which covers a broader range of action categories. They utilized both audio and video information to predict scores. Their method was validated on the Fis-V and FS1000 datasets, demonstrating its effectiveness across both datasets. Du et al. [26] used both textual and video information to train models for score prediction in figure skating. They collected a figure skating dataset, OlympicFS, which includes professional commentary annotations. Their method was validated on several figure skating datasets, achieving strong performance across all of them. For datasets without text annotations, they employed a transfer learning approach, first training on datasets with textual annotations and then fine-tuning on annotation-free datasets, resulting in improved performance. Zeng et al. [27] utilized RGB video, optical flow, and audio information for evaluation in rhythmic gymnastics and figure skating. Their method achieved strong performance on the Fis-V and Rhythmic Gymnastics datasets. Similarly, gedamu et al. [96] incorporated RGB video and semantic descriptions of actions as model inputs for action evaluation. Their approach also performed well on the MTL-AQA and Rhythmic Gymnastics datasets, further demonstrating the effectiveness of multi-modal methods. These studies highlight that by integrating information from different modalities, it is possible to capture the finer details and dynamic changes within the performance, thereby enhancing the accuracy and robustness of AQA tasks.

Multi-person interaction action assessment

Actions can be categorized into individual actions and multi-person interactive actions based on the number of participants. In the evaluation of individual actions, the focus is primarily on the athlete's personal performance. In contrast, for multi-person interactive actions, it is necessary to evaluate not only individual performance but also the dynamic interactions between participants and their impact on the overall performance. Gao et al. [34] collected a synchronized diving dataset called TASD-2 and proposed an AQA framework to evaluating interactive actions. The framework assessed both synchronization and execution scores on the TASD-2 dataset, using a multi-task learning approach to train the model. Finally, the proposed method achieved a SRCC of 0.89 on the TASD-2 dataset. Gao et al. [39] further explored the evaluation of interactive actions by collecting a dataset of synchronized figure skating videos, named PaSk. Building upon the method proposed by gao et al. [34], they introduced an automated model for evaluation. The Spearman correlation on the TASD-2 dataset improved by nearly two percentage, and reached 0.64 on the PaSk dataset.

Zhang et al. [38] collected a synchronized swimming dataset, LOGO, which presents unique challenges due to its nature as a team sport with longer performance durations, making it more complex compared to sports like diving or figure skating. They proposed an embeddable GOAT module that can be integrated into AQA models. The authors embedded this module into several existing AQA methods [22, 37, 49], and after embedding, these methods showed improved performance on the LOGO dataset. Furthermore, they validated the approach on several multi-person sports datasets, where embedding the GOAT module into the CoRe model resulted in performance improvements.

Fine-grained action assessment

Some traditional AQA methods typically evaluate actions based on coarse-grained action labels, whereas fine-grained AQA aims to capture subtle differences in actions through more precise temporal and spatial analysis, providing higher-precision evaluation results. Some researchers have attempted to model actions in a fine-grained manner to enhance both the accuracy and interpretability of the model's evaluation. Xu et al. [37] collected a diving dataset, FineDiving, which includes detailed annotations at both coarse-grained and fine-grained levels. They conducted experiments on this dataset using the previous methods [22, 49]. Additionally, they introduced a program-aware assessment method, which outperformed the aforementioned methods on the FineDiving dataset. Building upon this approach, [82] incorporated a spatial attention mechanism, and their proposed STSA model demonstrated further improvements in performance on the FineDiving dataset. [67] performed the AQA task using both the original videos and the videos after applying object detection. Additionally, they employed an unsupervised learning approach to segment the videos into multiple subaction clips. Their method demonstrated improvements across several AQA datasets in sports. Similarly, xu et al. [80] provided additional mask annotations for the FineDiving dataset, marking human-centered foreground action regions. They proposed the FineParser framework for fine-grained understanding of actions. By training the model with additional mask annotations, the FineParser framework focuses more on the athlete, leading to improved performance. Their method outperformed existing approaches on both the FineDiving and MTL-AQA datasets. Zhou et al. [88] defined the AQA task as a classification task from coarse to fine, simulating how a referee first determines a rough grade and then scores based on specific details. They proposed the CoFInAl framework, which was validated on two long-term action datasets. Additionally, they conducted experiments using various feature extractors to validate that their framework can effectively mitigate the overfitting problem commonly encountered in transfer learning. Ji et al. [40] collected a figure skating dataset, FineFS, which includes RGB videos, skeleton data, and detailed annotation information for each sample. They proposed LUSD-Net, a method designed to predict the PCS and TES for figure skating videos. Their approach outperformed several existing methods on the FineFS dataset and demonstrated strong performance in both short programs and long programs.

Interpretable feedback

In real-world sports settings, AQA systems are expected to function as practical decision-support tools rather than merely predictive models. Beyond producing final scores, such systems should deliver interpretable feedback that can be directly utilized by athletes, coaches, and judges to understand performance quality, identify technical deficiencies, and support training or evaluation processes. However, many existing AQA systems remain difficult to deploy in practice due to their limited transparency and user accessibility. This gap has motivated recent research to explore how AQA systems can be applied to provide interpretable, user-centered feedback and to assess their effectiveness through real-world usage and stakeholder-oriented evaluations.

Matsuyama et al. [92] presented IRIS, an AQA system designed to provide interpretable feedback for figure skating by aligning model outputs with official judging rules. Beyond predicting overall scores, IRIS generates criterion-level sub-scores that correspond to different aspects of the performance, enabling users to better understand how the final evaluation is formed. In addition to benchmarking the system on the MIT-Skating dataset, the authors emphasized the importance of assessing interpretability from a practical standpoint. They developed a visualization interface for IRIS outputs and conducted a user study involving figure skating athletes. The survey results indicated that the generated feedback was highly consistent with the participants' expectations of professional judging, highlighting the system's practical value in training and self-assessment.

Similarly, Okamoto et al. [95] proposed the NS-AQA system to support transparent evaluation in diving. The system produces structured, rule-consistent analysis by decomposing diving performances into multiple phases and generating detailed reports based on interpretable attributes such as body posture and movement configuration. Rather than focusing solely on predictive accuracy, the authors evaluated NS-AQA by presenting its outputs to professional stakeholders. Experiments involving six certified judges across 50 diving evaluations showed that 90% of the system's assessments were approved, while experienced diving coaches further confirmed the usefulness of the generated feedback for performance analysis and training support.

Beyond structured scoring explanations, recent work has also explored more expressive forms of feedback. Zhang et al. [51] introduced Narrative Action Evaluation (NAE), which extends conventional AQA by generating expert-style textual comments in addition to numerical scores. By providing natural-language explanations, the system helps users better understand performance strengths and weaknesses in an intuitive manner. From an application perspective, such narrative feedback can facilitate communication between automated systems and human users, particularly in coaching and educational settings. Experimental results demonstrated that the proposed framework not only produces high-quality expert comments but also improves overall evaluation performance compared to traditional score-only AQA approaches.

In another application-oriented study, Majeedi et al. [98] proposed the RICA^2 framework, which emphasizes uncertainty-aware and interpretable scoring. The system outputs not only predicted action quality scores but also associated uncertainty estimates, allowing users to assess the reliability of the evaluations. This capability is particularly valuable in real-world deployment, where ambiguous or borderline performances are common. The authors validated the framework across multiple datasets and introduced

Kendall's tau (τ) to quantify the relationship between prediction uncertainty and error, further supporting the practical interpretability of the system outputs.

In real-world sports settings, AQA systems are expected to function as practical decision-support tools rather than merely predictive models. Beyond producing final scores, such systems should deliver interpretable feedback that can be directly utilized by athletes, coaches, and judges to understand performance quality, identify technical deficiencies, and support training or evaluation processes. However, many existing AQA systems remain difficult to deploy in practice due to their limited transparency and user accessibility. This gap has motivated recent research to explore how AQA systems can be applied to provide interpretable, user-centered feedback and to assess their effectiveness through real-world usage and stakeholder-oriented evaluations.

Application of AQA in medical care

In the medical field, AQA technology is mainly applied in physical rehabilitation and abnormal motion detection. In physical rehabilitation, AQA helps to automatically assess the accuracy and consistency of patient movements during recovery, which supports healthcare professionals in developing more effective and evidence-based treatment plans. Abnormal motion detection focuses on identifying movement patterns associated with medical conditions to assist in clinical diagnosis and assessment. This section introduces the practical development of AQA applications in these two representative scenarios. The overall structure of the chapter is shown in Figure 9.

Physical rehabilitation

Physical rehabilitation aims to help patients regain motor function. AQA technologies can objectively evaluate movement quality, assisting clinicians in developing or refining rehabilitation plans and ultimately improving recovery efficiency. Several researchers have collected a series of relevant datasets to provide data support for related studies. Vakanski et al. [18] compiled the UI-PRMD dataset, which includes physical rehabilitation movements performed by 10 healthy participants. These samples were captured using Vicon optical trackers and Microsoft Kinect sensors. However, this dataset is limited to movements performed by healthy participants, lacking data on actions performed by unhealthy patients. In contrast, Capecci et al. [19] collected the KIMORE dataset,

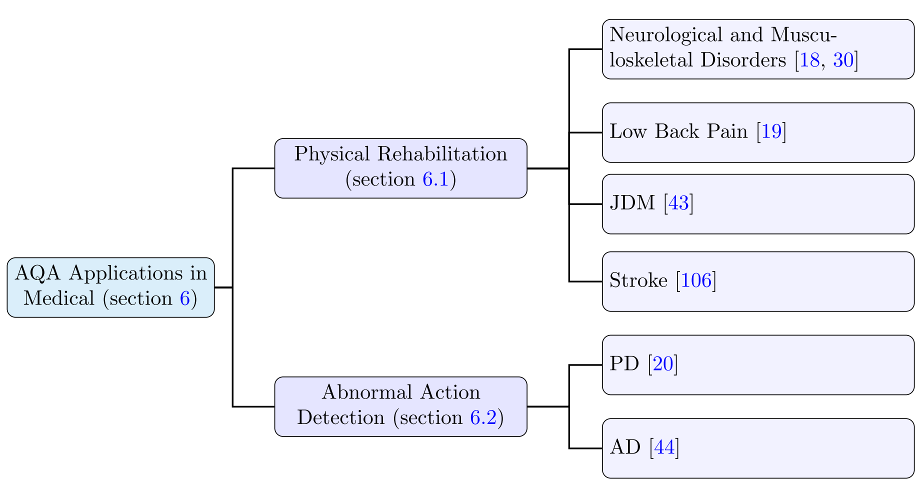

Fig. 9 Overview of AQA applications in medical care

which focuses on rehabilitation exercises for low back pain. This dataset includes samples from healthy participants and patients with movement impairments, encompassing three modalities: RGB video, depth video, and skeleton data. Zhou et al. [43] collected the JDM dataset and proposed an AQA method to assess muscle strength in patients with JDM disease. By integrating video-based augmented reality technology, the AQA output is visualized through a virtual avatar, helping medical professionals analyze the condition of JDM patients. To evaluate the effectiveness of the system in practical applications, a user study was conducted at the Children's Hospital of the Capital Institute of Pediatrics in Beijing, China. Several non-pediatric clinicians used the system to assess muscle strength of patients with JDM. With the help of this system, their evaluation efficiency and accuracy improved significantly.

The evaluation of physical rehabilitation training is predominantly based on subjective assessment scales [108]. This approach relies heavily on the expertise of the evaluator, which may lead to inconsistencies in the assessment results. To address this, Li et al. [30] developed a multi-modal dataset named FineRehab, focusing on rehabilitation training for neurological and musculoskeletal disorders. They proposed a novel rehabilitation motion assessment framework to annotate each sample. The framework evaluates motions from three dimensions: completeness, correctness, and smoothness, using a scoring scale of 0 to 4. Each motion segment is rated by multiple annotators, and the final score is determined through a majority voting mechanism. The consistency of evaluations among annotators is validated by calculating the Intraclass Correlation Coefficient (ICC), ensuring the reliability of the results. Scores for each dimension are weighted according to expert-defined criteria, and the final score is obtained through a weighted summation.

Most evaluation models for physical rehabilitation exercises are based on skeletal data. These methods typically rely on specialized equipment to capture high-precision motion data, making them widely applicable in hospitals or professional rehabilitation centers equipped with advanced instruments and technical support. However, their applicability to the general population is limited, as such specialized equipment is rarely available in ordinary households or non-professional settings. Kim et al. [106] evaluated stroke survivors' grasping actions using two modalities: RGB videos and optical flow. To reduce the burden of frequent hospital visits, they developed a self-administered rehabilitation system for stroke survivors. The system enables users to upload videos of their grasping actions, which are assessed by an AQA model. Based on the assessment scores, the system recommends suitable rehabilitation exercises to support patients in their recovery process. To validate the system's effectiveness, 10 patients, 10 physiatrists, and 10 therapists were invited to evaluate it using the Mobile Application Rating Scale (MARS). While the system received widespread recognition for its quality application, its performance in subjective application was rated lower.

Abnormal action detection

In addition to its widespread use in rehabilitation, AQA technology is also used to detect abnormal movement patterns, helping to early diagnose of motor-related disorders such as AD and PD. Bruce et al. [44] collected the EHE dataset, which consists of skeletal motion data captured from elderly participants, including healthy individuals and AD patients. They developed a 2T-GCN framework based on Graph Convolutional

Networks (GCN) to detect anomalous skeletal samples and evaluate motion scores. The proposed method was validated on both the UI-PRMD and EHE datasets. It demonstrated the ability to predict abnormalities in non-gait-related motions, providing support for AD diagnosis. Furthermore, the numerical scores generated were positively correlated with the severity of AD. Similarly, Dadashzadeh et al. [20] collected a dataset for PD that includes tasks performed by patients with PD.

Liu et al. [23] proposed a GCN-based EGCN framework that takes advantage of positional and directional streams of skeletal data to detect abnormal motions in skeletal samples. Their approach introduced four different integration strategies, which were validated on the UI-PRMD and KIMORE datasets, achieving significant improvements compared to other unimodal and integrated methods. Building on this, Bruce et al. [72] introduced new integration strategies in an enhanced framework, EGCN++, which was evaluated on the UI-PRMD, KIMORE, and EHE datasets, demonstrating further performance improvements over the original EGCN framework. Yao et al. [24] developed a performance metric network to assign score labels to the UI-PRMD dataset, which lacks action score annotations. This network uses classification annotations (healthy or unhealthy) as labels to quantify differences in movement patterns between healthy and unhealthy skeletal sequences, converting these differences into action evaluation scores. They evaluated the separability of the generated scores to validate the effectiveness of the performance metric network. Additionally, they proposed an AQA network that utilizes skeletal data and contrastive learning to predict action scores. The network was validated on both the UI-PRMD and KIMORE datasets. For the UI-PRMD dataset, the action score labels were generated by the performance metric network, while the KIMORE dataset inherently provides action score annotations.

Application of AQA in skill assessment

In the field of skill assessment, AQA has proven effective not only in evaluating simple daily activities such as using chopsticks or braiding hair, but also in assessing complex professional skills such as surgery and musical performance. This section provides a comprehensive review of the research and applications of AQA in skill assessment, focusing on both daily life skills and specialized professional skills. To help readers better understand the structure of this section, an overall structure is presented in Figure 10.

Daily life skills

In the domain of daily skill assessment, Doughty et al. [11] introduced the EPIC Skills dataset to support skill pair ranking tasks. The dataset includes a variety of basic daily

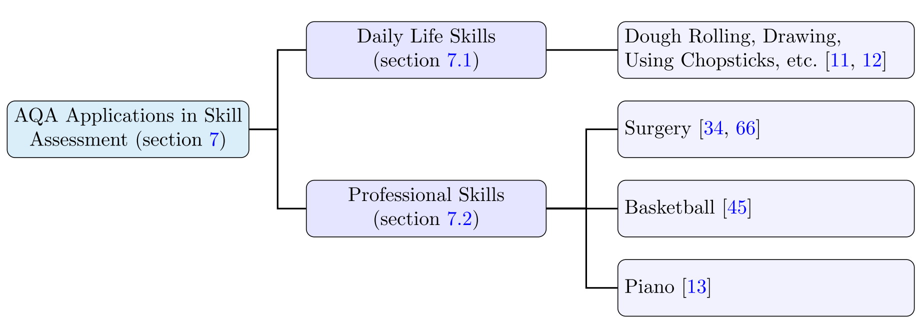

Fig. 10 Overview of AQA applications in skill assessment

activities such as dough rolling, drawing and using chopsticks, as well as a more complex surgical task derived from the JIGSAWS dataset collected by Gao et al. [21]. In their subsequent work, Doughty et al. also proposed the BEST dataset [12], which covers five common categories of daily skills, further expanding the foundation for research in this field.

In model design, Doughty et al. [11] proposed a pairwise deep ranking model to compare skill levels between two videos. The model achieved an average pairwise accuracy of approximately 0.76 on the EPIC Skills dataset, demonstrating strong performance and generalization across various skill ranking tasks. To further improve ranking performance, Doughty et al. [12] introduced an attention mechanism and developed a rank-aware loss function to guide the attention module to features relevant to skill ranking. This enhanced approach achieved pairwise accuracies of 0.803 and 0.812 in the EPIC Skills and BEST datasets, respectively, significantly outperforming earlier methods. In addition, Pan et al. [69] proposed an adaptive action assessment method, initially designed for regression tasks in sports-related actions but also applicable to pairwise skill ranking. On both the EPIC Skills and BEST datasets, this method outperformed several previous approaches, demonstrating strong adaptability. More recently, Zhang et al. [70] introduced the AdaST framework, which leverages the similarity between different skills to enhance the ranking process. AdaST achieved pairwise accuracies of 0.8832 and 0.8534 on the EPIC Skills and BEST datasets, respectively, marking a substantial improvement in skill assessment performance.

Professional skills

In the field of professional skill assessment, surgical skill evaluation has attracted considerable research attention. Liu et al. [46] collected video data from laparoscopic gastrectomy procedures and proposed a method that combines handcrafted features with high-level semantic representations extracted by deep learning for automatic surgical skill assessment. The method was empirically validated on the dataset, where both junior and senior surgeons provided performance ratings. Model performance was measured using the SRCC between predicted scores and human annotations. The results showed that the model achieved scoring performance comparable to that of junior surgeons and was capable of producing frame-level weighted score sequences, which enhanced interpretability of the model. However, as the method was validated only on a single dataset, its generalizability remains to be further investigated. In their subsequent work, Liu et al. [31] collected a clinical dataset covering various types of laparoscopic surgeries and proposed a multipath surgical skill assessment framework. This framework models the surgical procedure from multiple perspectives, including instrument usage patterns and surgical event recognition, while also capturing the dependencies among different analytical paths. The method achieved strong performance on both the JIGSAWS and the clinical datasets. However, although these datasets were collected from real clinical environments and have substantial practical value, they remain unavailable to the public due to ethical concerns such as protecting patient privacy, which limits further research in this area.

In the execution of surgical skill, the motion of the tip of the surgical tool can be considered the primary information, while the motion at the handle is relatively less important and can be viewed as secondary information. Gao et al. [34] addressed this asymmetric interaction between primary and secondary information by extracting AIM features and integrating them with global video features to perform the AQA task. Their method achieved strong performance on the JIGSAWS dataset. The evaluation of any action involves uncertainty due to its internal complexity and the subjectivity of the scoring, and the surgical skill assessment is no exception. The UD-AQA model designed by Zhou et al. [63] simulates this uncertainty by generating diverse scores. This method has not only shown promising results in the sports domain, but has also demonstrated strong performance in the JIGSAWS dataset for surgical skill assessment. In contrast, zhang et al. [66] introduced an auto-embedding DAE module that encodes features as Gaussian distributions to simulate uncertainty. This approach, when embedded into CoRe and tested on the JIGSAWS dataset, showed lower results compared to UD-AQA.

In addition to surgical skills, some studies have explored the assessment of other types of professional skills. For example, Bertasius et al. [45] proposed a video analysis approach based on a first-person perspective to evaluate basketball performance. Their method used 250 labeled pairs of player videos to predict which player performed better and model performance was evaluated using prediction accuracy. The results showed that the model could reliably rank player performance, even in the absence of explicit quantitative scores. Parmar et al. [13] focused on evaluating piano performance, integrating video and audio modalities. Their model achieved high evaluation accuracy on the PISA dataset they developed.

Challenges and future work

In this section, we will provide a detailed analysis of the current challenges faced by AQA, and discuss potential directions for future work.

Challenges

AQA is a complex and challenging research field that involves a range of technical and application-level issues. AQA systems must handle diverse video data, address subjective evaluation standards, and possess strong generalization capabilities to adapt to various application scenarios. As AQA technology is applied to fields such as sports, medical rehabilitation, and skill assessment, each area presents unique requirements, leading to distinct challenges in specific domains.

Camera-related challenges. RGB video data are often affected by various factors during the recording process, leading to noise in the data, which negatively impacts model performance. This issue is particularly prevalent in AQA datasets within the domain of sports. For example, improper camera angles may result in certain body parts being obscured or hidden. Additionally, complex or extreme movements, such as diving, where athletes perform rapid limb movements, often cause motion blur at the edges of the body, further exacerbating noise interference. These factors pose significant challenges for AQA models in accurately recognizing and analyzing movements.

Data annotation challenges. To ensure the accuracy of data annotations, annotators typically require specialized knowledge, making the annotation process for AQA datasets both time-consuming and relatively costly. For example, Figure 11 presents sample images from the JIGSAWS dataset, showcasing three highly specialized actions: suturing, knot tying, and needle passing. Individuals without relevant expertise may even be unable to recognize what actions are being performed. Furthermore, for some complex

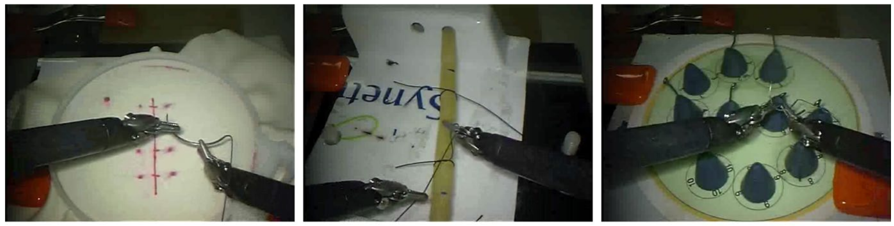
(a) Suturing (b) Knot Tying (c) Needle Passing

Fig. 11 Examples of JIGSAWS datasets

models, more detailed annotation information may be required. For example, the TSA model proposed by xu et al. [37] necessitates fine-grained segmentation of actions to identify individual subactions. As a result, in the FineDiving dataset, they not only annotated the overall temporal boundaries of the actions but also meticulously recorded the starting and ending frames for each subaction. This additional annotation work further increases the complexity and cost of data labeling.

Multi-modal data challenges. In recent years, multi-modal methods have gained increasing attention in the field of AQA, and the development of these methods relies heavily on high-quality multi-modal data. Many studies have utilized audio and video information to perform AQA tasks, with audio often being extracted from video. However, for other multi-modal approaches, such as those combining text and video information, a dataset containing commentary information, OlympicFS, was collected by du et al. [26] for figure skating. Additionally, there are multi-modal methods that utilize skeleton and video information. While skeleton data is relatively easier to collect in the medical rehabilitation domain, it presents significant challenges in the sports domain, particularly for high-intensity actions. High-speed motion often leads to substantial errors in skeleton data, making such data relatively scarce in the sports domain.

Data privacy challenges. As data collection deepens, data privacy issues have become significant obstacles in the data collection process. This is particularly true in the medical rehabilitation domain, where the collected data often involves patients' physical conditions and personal health information. The collection and use of such data must adhere to strict ethical guidelines and legal requirements. As a result, in the medical care domain, most datasets take the form of skeleton data, while RGB video data is relatively scarce. In the fields of medical care and skill assessment, the datasets shown in Tables 2 and 3 are not only limited in number but also restricted by privacy concerns, with many datasets not publicly available. This situation presents a challenge for researchers who plan to pursue AQA tasks in these fields in the future.

Methods challenges. The execution process of actions varies in duration. Currently, many AQA models have achieved strong performance on short-term actions. However, for long-duration actions, such as figure skating and surgery, which involve more intricate and complex information, it is necessary to model the temporal dynamics within large video sequences. Existing AQA models demonstrate limited performance in capturing long dependencies, and there is still a need for continuous improvement in this regard. Currently, most AQA methods map input action videos to corresponding scores, with only a few methods offering explanations of the evaluation process. These methods lack interpretive feedback on the scoring process, leaving users unaware of the rationale behind the scores. Interpretive feedback helps users understand the model's decision-making process, enabling better judgments and adjustments. Therefore, improving model interpretability and providing such feedback is crucial.

Future work

Data expansion. Existing AQA datasets are highly imbalanced, with most publicly available datasets focusing on the sports domain, while datasets from other fields are relatively scarce. This limits the development of AQA in broader application contexts. Therefore, future work should address the gap by supplementing datasets from the medical care and skill assessment domains. To advance multi-modal AQA technologies, future research should also aim to include datasets from additional modalities. Due to the complexity and high speed of sports movements, skeleton data is challenging to obtain. In this domain, skeleton data collection can follow the approach proposed by nekoui et al. [100, 102], which uses pose estimation models for extreme actions to extract skeleton data of athletes from RGB videos, thereby augmenting skeleton data in the sports domain. In the medical care domain, privacy concerns severely limit the availability of public datasets, and most available datasets consist of skeleton data. To foster the development of AQA technologies in this field, future efforts should focus on strengthening collaboration with medical care institutions, promoting the sharing of more anonymized and de-identified data to address the data scarcity. For obtaining textual modality data, the approach used by majeedi et al. [98] can be applied, where large language models are utilized to generate textual descriptions of actions, thereby supplementing the relevant textual data.

Unsupervised and semi-supervised learning. Unlike datasets from other fields, annotating AQA datasets requires significant time and resources, as well as a high level of expertise, which is particularly evident in the medical care and skill assessment domains. Works proposed by zhang et al. [50] and yun et al. [85] have utilized unsupervised and semi-supervised learning to address the AQA task. The semi-supervised teacher-student architecture proposed by yun et al. [85] outperforms some fully supervised learning methods, even when only 50% of the data is labeled. The combination of unsupervised and semi-supervised learning reduces the cost of data annotation while enabling the model to learn useful features even in the absence of large amounts of labeled data. Therefore, this approach holds significant potential and warrants further development in AQA tasks.

Expanding practical applications. Most AQA methods are still limited to validation on benchmark datasets. While these methods perform well on standard datasets, research into their application in real-world scenarios is still lacking. To promote the practical application of AQA technologies, future work should shift from benchmark testing to real-world application scenarios. The challenges in real-world applications are much greater than those found in datasets, as they involve more complex and dynamic environments and tasks. Some researchers have already applied AQA technologies to real-world applications. For example, the figure skating assessment model proposed by matsuyama et al. [92] generates detailed scores for each subaction, providing more comprehensive feedback. The evaluation of AQA methods can draw on the method proposed by matsuyama et al. [92] and okamoto et al. [95], incorporating user feedback and expert suggestions from relevant fields to drive the practical adoption of AQA methods. Additionally, future research can further integrate AQA technologies with other emerging technologies, such as virtual reality and augmented reality. Similar to the work of zhou et al. [43], augmented reality could be used to visualize AQA outputs as virtual characters, assisting medical care professionals and other experts in real-time diagnostics and feedback.

Narrative action evaluation. Currently, score-based AQA methods have achieved significant performance improvements, effectively quantifying action quality. However, with the advancement of AQA technology, future research should not be limited to providing a single score, but should further expand into more comprehensive and nuanced forms of feedback to help users better understand the model's decision-making process. In addition to scores, AQA technology should also generate detailed feedback, explaining the rationale behind the evaluation results, providing specific assessments and recommendations for each action, and enhancing user understanding and trust. Zhang et al. [51] introduced the Narrative action evaluation task, which aims not only to assess the quality of actions but also to generate professional evaluation reports using natural language, highlighting strengths and weaknesses in action execution. Compared to traditional score-based evaluation methods, Narrative Action Evaluation offers stronger interpretability and guidance. It has the potential to become a key research direction in AQA, promoting the widespread practical application of the technology.

Limitation and conclusion

Limitation

While this survey aims to provide a systematic overview of AQA methods, several limitations should be noted. First, due to the rapid evolution of the field, some very recent studies-particularly in emerging areas such as multi-modal learning and data-efficient training-may not be fully covered. Second, the proposed taxonomy organizes methods according to their dominant methodological characteristics; however, many approaches naturally combine multiple paradigms, which may lead to unavoidable overlaps between categories. Finally, this review primarily focuses on methodological developments and does not extensively discuss practical deployment considerations, such as efficiency, scalability, or ethical issues.

Conclusion

A cross-domain examination of AQA reveals that, although AQA shares a common objective of evaluating human action performance, application-specific task requirements play a critical role in shaping model design and evaluation strategies. In sports scenarios, particularly competition scoring, AQA systems prioritize scoring accuracy and consistency with expert judgment, which often motivates the use of global or stage-aware temporal aggregation mechanisms to capture overall execution quality. In training-oriented sports applications, the demand for interpretable feedback further encourages fine-grained temporal modeling and the incorporation of semantic or phase-level representations. In medical care, including physical rehabilitation and abnormal action detection, robustness to inter-subject variability, reliability, and clinical interpretability are essential, leading to model designs that emphasize stable representations, explicit temporal alignment, and explainable assessment outputs. In contrast, skill assessment tasks, ranging from daily life activities to professional skills, place greater emphasis on generalization across diverse contexts, data efficiency, and, in some cases, real-time feedback, which promotes lightweight architectures, modular representations, and transferable evaluation criteria.

These domain-specific requirements directly influence AQA model design choices, including temporal modeling strategies, supervision granularity, and output forms. Rather than pursuing a uniform modeling paradigm, effective AQA systems are typically application-driven, with architectures and learning objectives tailored to the practical constraints and expectations of the target domain. Recognizing these distinctions is therefore crucial for developing AQA models that are not only accurate but also interpretable, robust, and deployable in real-world scenarios.

The goal of AQA tasks is to automatically assess the quality of an individual's performance in specific actions, with wide-ranging applications in sports, medical care, and skill assessment. In this review, we systematically examine the research progress in AQA, with a particular focus on deep learning-based approaches. We summarize commonly used datasets across different application domains and provide a detailed review of representative AQA methods in sports, medical care, and skill assessment. This survey aims to offer a comprehensive overview of recent advances in AQA, while providing theoretical insights and practical references to support future AQA model development and application.

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Author contributions
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Data availability
Data is provided within the manuscript.

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
Not applicable.

Code availability
The code used during the current study are available from the corresponding author on reasonable request.

Competing interests
The authors declare no competing interests.

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